CORE CONCEPTUAL MODEL

AI Data Centre Facilities Assurance Framework

Version 1.0

Canonical meanings, reasoning structure and controlled vocabularies for capability-centred AI infrastructure assurance

Artifact ID: AIDCF-002
Short code: CCM
Artifact type: Canonical foundation (normative)
Status: Draft for controlled review
Author: Siva Sethumadhavan

This document defines what AIDCF means by its core terms, how an AIDCF assurance argument is structured, which controlled codes every AIDCF artifact, dataset and tool shall use, and how the many level scales used across the framework relate to each other. It is second in precedence only to the AIDCF Manifesto (AIDCF-001).

Document control

Field	Value
Title	AIDCF Core Conceptual Model
Public ID	AIDCF-002
Short code (requirement IDs)	CCM
Version	1.0
Date	2026-10-10
Status	Draft for controlled review
Author	Siva Sethumadhavan
Artifact type	Canonical foundation (normative)
Depends on	AIDCF-001 AIDCF Manifesto
Downstream users	AIDCF-003 to AIDCF-019, AIDCF-VR-001, AIDCF-WP-001, the canonical registries and any conformant tooling
Machine-readable companion	`registries/vocabularies/*.csv` (controlled vocabularies defined in Part VI of this document)
Intended audience	Methodology owners; AI infrastructure and data-centre architects; electrical, mechanical, controls and network engineers; assessors and reviewers; decision authorities; ontology and tooling engineers
Publication boundary	Independent methodology. Not a substitute for law, licensed engineering, safety approval, regulatory certification or vendor design authority.

Revision history

Version	Date	Summary
1.0	2026-10-10	First issue of the canonical Core Conceptual Model: assurance chain, 15026-2/GSN mapping, core concept definitions, MATERIAL, control model, controlled vocabularies, gate and score doctrine summaries, ladders crosswalk, artifact register, legacy domain mapping, interpretation rules and glossary. Section 36: an applicable NOT_ASSESSED gate applies its AIDCF-010 UNKNOWN ceiling (aligned with AIDCF-010 and AIDCF-016).

Contents

- Part I – Authority, precedence and conformance
- Part II – The assurance problem and design stance
- Part III – The canonical assurance chain and assurance-case structure
- Part IV – Core concepts
- Part V – Materiality and the control model
- Part VI – Controlled vocabularies
- Part VII – Gate, decision and score doctrine (summary)
- Part VIII – Ladders crosswalk
- Part IX – Artifact register and precedence graph
- Part X – Legacy eight-domain to fifteen-domain mapping
- Part XI – Interpretation rules
- Part XII – Conformance requirements index
- Appendix A – Glossary
- Appendix B – AIDCF-014 class coverage

Part I – Authority, precedence and conformance

1. Purpose

AIDCF-001 states what AIDCF is for and what it will not do. This document states what AIDCF's words mean and how its reasoning is assembled. It exists so that an electrical engineer reviewing UPS autonomy, a network architect reviewing a scale-out fabric, an assessor writing a Finding and a developer writing a SHACL shape all use the same object, the same code and the same rule for the same thing.

The Core Conceptual Model (CCM) has five jobs:

1. Fix the **canonical assurance chain**, the single ordered structure from Decision Question to Assurance Decision, and map it to the established assurance-case vocabulary of ISO/IEC/IEEE 15026-2 and Goal Structuring Notation (GSN).
2. Define every **core concept** with its attributes, invariants and the AIDCF-014 object class that carries it.
3. Define **MATERIAL** and the **control model**, the two ideas on which most disagreements in assurance work turn.
4. Publish the **controlled vocabularies**: claim types, Assurance Levels, control results, gate results, decision states, confidence, evidence classes and grades, UNKNOWN type and severity, operating states, methods, control types, lifecycle stages and score bands, with legacy aliases.
5. Provide a **ladders crosswalk** so that the level scales defined across AIDCF artifacts are never confused with each other, and state which of them are normative for decisions.

2. Scope and exclusions

In scope: canonical meanings; the structure of the assurance argument; controlled codes and aliases; precedence and identifier rules; interpretation rules that bound every AIDCF conclusion.

Out of scope, and owned elsewhere:

Subject	Owner
Entity taxonomy, facets, relationship semantics, capacity and redundancy semantics	AIDCF-003
Threat, hazard, failure-mode and scenario taxonomy and analysis method	AIDCF-004
Mapping to AI Trust Graph	AIDCF-005
Constitutional assurance principles (APS-001 to APS-058)	AIDCF-006
The 360 controls, their criteria, evidence profiles and gate mappings	AIDCF-007A
Evidence metadata, grading, sufficiency and conflict workflow	AIDCF-008
Assessment procedures and workpapers	AIDCF-009
Gate criteria, roll-up, decision ceilings and score computation	AIDCF-010
Programme models, Assurance Level profiles and stage gates	AIDCF-011
Report classes and statement wording	AIDCF-012
Reference architecture layers	AIDCF-013
Object envelopes, relationship registry, serialisation and validation	AIDCF-014 and AIDCF-016
Maturity model	AIDCF-017

3. Precedence

AIDCF artifacts are read in the following order of authority. Where two artifacts conflict, the higher one governs until the conflict is resolved through methodology change governance and recorded as an AIDCF-014 BaselineIssue.

1. AIDCF-001 AIDCF Manifesto (purpose, identity, commitments).
2. **AIDCF-002 Core Conceptual Model (this document: meanings, chain, vocabularies).**
3. AIDCF-003 Infrastructure Ontology and Taxonomy.
4. AIDCF-004 Threat, Hazard and Failure Model.
5. AIDCF-005 AIDCF–AI Trust Graph Integration Standard.
6. AIDCF-006 Assurance Principles Standard.
7. Downstream standards AIDCF-007A to AIDCF-019, each authoritative for its own subject (Section 2) and subordinate to items 1–6 for meaning.
8. Validation, whitepaper and reference implementation material (AIDCF-VR-001, AIDCF-WP-001, AIDCF-015): informative unless they quote a normative source.
9. Tooling: declared implementation only, never hidden authority.

Applicable law, licensed engineering judgement and the authority having jurisdiction over life safety prevail over every AIDCF artifact (AIDCF-005 safety-conflict rule; AIDCF-006 non-substitution).

> **CCM-REQ-001 Precedence.** Where an AIDCF artifact, registry or tool uses a term, code or rule defined in this document with a different meaning, the meaning in this document shall govern and the divergence shall be recorded as a BaselineIssue until resolved.

> **CCM-REQ-002 Subject ownership.** A downstream artifact may specialise or strengthen a rule owned by a higher artifact but shall not weaken, redefine or silently reinterpret it.

4. Normative language

Keyword	Meaning
shall / **shall not**	Requirement. Conformance is not possible without satisfying it.
should / **should not**	Recommendation. A departure is permitted only with a recorded rationale.
may	Permission.
can	Capability or possibility; not a requirement.

This usage follows ISO/IEC Directives Part 2. Text marked *informative*, notes and examples are not requirements. Tables that list controlled codes are normative unless marked *illustrative*.

The defined term **MATERIAL** (Section 22) is normative wherever it appears in an AIDCF artifact, whether capitalised or not.

5. What conformance to this document means

Three kinds of subject can claim conformance to AIDCF-002:

Subject	Conforms when
An AIDCF artifact	It uses the chain step names, concept names and controlled codes defined here, cites other artifacts by public ID, and contains no conflicting definition.
An assessment or assurance dataset	Every record uses canonical codes, every material claim sits in a chain that can be traced from Decision Question to Assurance Decision, and the interpretation rules in Part XI are honoured.
A tool or registry	It stores and exchanges canonical codes (or declared aliases mapped on ingest), preserves UNKNOWN, and never computes a decision state without a recorded accountable approval.

Requirements are numbered CCM-REQ-001 onwards in order of appearance and indexed in Part XII.

> **CCM-REQ-003 Canonical codes.** Artifacts, datasets and tools shall use the canonical codes in Part VI. Legacy codes may be accepted on ingest only through the published alias tables and shall be stored in canonical form with the legacy value retained as provenance.
>
> **CCM-REQ-004 Public identifiers.** Artifacts shall be cross-referenced only by public ID (Part IX), never by ordinal phrases such as "Artifact 4" or by superseded internal IDs.

Part II – The assurance problem and design stance

6. The problem

An AI capability such as a distributed-training cluster or a latency-bound inference service is delivered by a stack that crosses at least four professional boundaries: facility engineering (power, cooling, fire, structure), operational technology (BMS, EPMS, CDU and plant controllers), AI infrastructure (accelerators, scale-up and scale-out

fabrics, parallel storage, schedulers) and operations and supply (people, procedures, OEMs, utilities). Each discipline has mature standards, and each can produce a valid review that is silent on the failure paths that matter most, because those paths cross the boundary between disciplines. Typical examples:

- A 2N UPS design whose two paths share one EPMS, one battery-monitoring vendor, one maintenance contractor and one static-switch firmware baseline.
- A direct-to-chip liquid-cooled hall that is adequately rated at steady state but whose ride-through during a chiller restart or CDU pump failover is shorter than the time the scheduler needs to checkpoint and drain jobs.
- A checkpoint tier that survives storage-node loss but sits in the same electrical and thermal fault domain as the compute it protects.
- A management network that is logically segregated but reachable from a remote-support path whose credentials can also operate cooling plant.

Three further properties make the problem hard. **Concentration**: a small physical failure can disable a disproportionately valuable capability. **State dependence**: exposure in maintenance, degraded and recovery states differs from normal operation, and those are precisely the states that evidence rarely covers. **Evidence decay**: high-density AI estates change hardware, firmware, topology and thermal load faster than traditional facility documentation is revised.

7. Design stance

AIDCF responds with eight design commitments. Each is binding through the parts of this document noted.

Commitment	Meaning	Where enforced
Capability first	The object of assurance is a bounded Capability Instance and its Service Outcomes, not a building or an asset list.	Parts III–IV
Argument, not checklist	AIDCF produces an assurance case: a structured argument from evidence to a bounded top claim. Controls are the testable requirements inside that argument.	Part III, Section 23
Claim depth is explicit	Design, implementation, operation, effectiveness and resilience are different claims needing different evidence.	Sections 23.3, 24
Evidence-gated	No positive result without evidence at or above the profile floor; insufficient evidence yields UNKNOWN, never a discounted score.	Sections 26, 30
Non-compensating gates	Fifteen Critical Gates protect prerequisites that averages must not hide.	Part VII
UNKNOWN is preserved	Missing, stale, conflicting or inaccessible information is recorded, typed and given a severity with a defined decision effect.	Sections 20.3, 31
Bounded conclusions	Every conclusion is bounded by scope, operating states, time and evidence.	Part XI
Human authority	Engines compute and recommend; accountable people decide and approve.	Sections 21.5, 36

Part III – The canonical assurance chain and assurance-case structure

8. The top claim

Every AIDCF assurance case has exactly one top claim, written in this form:

> **"Capability Instance X can achieve Service Outcomes Y within tolerances T under declared operating states S, for Decision Question Q, as of date D."**

Each placeholder is mandatory and each narrows the claim:

Placeholder	Bound	Example (illustrative)
X	Which deployment	"Training Cluster TC-2, Hall 3, Site A, architecture baseline AB-2026-07"
Y	Which results	"Job-progress availability; checkpoint durability; restart within objective"
T	How much deviation is tolerable	"No more than one unplanned interruption per quarter; checkpoint loss bounded to the last completed interval; restart within the declared recovery objective" (all values profile parameters)
S	In which conditions	NORMAL, MAINTENANCE (one CDU out of service), DEGRADED (one utility feed lost), RECOVERY
Q	For what decision	"Is TC-2 ready to accept a production training contract?"
D	When	Assessment as-of date and evidence period

> **CCM-REQ-005 Top claim.** Every assessment shall state a top claim containing a Capability Instance, one or more Service Outcomes with tolerances, the declared operating states, the Decision Question and the as-of date. An assessment without all six elements shall not issue a decision state other than INCONCLUSIVE.

9. The sixteen-step chain

The canonical assurance chain is the **only** reasoning chain in AIDCF. Earlier variants (the Manifesto §14 reasoning chain, the AIDCF-003 chain, the AIDCF-014 chain and the README chain) are summaries of it and shall be read as such.

```
 1 Decision Question
 2 Capability Instance
 3 Service Outcome
 4 Required Function
 5 Architecture (InfrastructureSystem / Asset / Interface)
 6 Dependency and Fault Domain
 7 Threat / Hazard / Failure Scenario
 8 Safeguard
 9 Control
10 Evidence Requirement
11 Evidence Object
12 Control Assessment (result)
13 Finding / UNKNOWN Record
14 Gate Result
15 Score Snapshot and Decision Recommendation
16 Assurance Decision
   └─ Assurance Report and Reassessment Trigger close the loop back to step 2
```

Step	Name	Question answered	Primary AIDCF-014 classes	Detail owner	AIDCF-014 trace link
1	Decision Question	What decision will rely on this assessment, and who will make it?	Assessment (decisionQuestion), AssuranceProgram	AIDCF-011	TR-15
2	Capability Instance	Which bounded deployment is the subject?	CapabilityInstance	AIDCF-003	TR-01
3	Service Outcome	What must it achieve, within which tolerances?	ServiceOutcome	AIDCF-003	TR-01, TR-02
4	Required Function	Which functions must be delivered for each outcome?	RequiredFunction	AIDCF-003	TR-02, TR-03
5	Architecture	Which systems, assets and interfaces implement each function, as built?	ArchitectureSnapshot, InfrastructureSystem, AssetInstance, Interface, Resource, CapacityAssertion	AIDCF-003, AIDCF-013	TR-03, TR-04
6	Dependency and Fault Domain	What does each element rely on, and what fails together?	Dependency, FaultDomain, OperatingState	AIDCF-003	TR-04, TR-05
7	Threat / Hazard / Failure Scenario	How can the outcome be defeated?	Threat, Hazard, FailureMode, FailureScenario	AIDCF-004	TR-05, TR-06
8	Safeguard	Which real measures prevent, detect, contain or recover?	Safeguard	AIDCF-004	TR-06, TR-07
9	Control	Which AIDCF requirements test those safeguards?	ControlDomain, ControlFamily, Control	AIDCF-007A	TR-07, TR-08
10	Evidence Requirement	What evidence would demonstrate each control at the required claim type?	EvidenceRequirement	AIDCF-008	TR-08, TR-09
11	Evidence Object	What evidence was actually obtained?	EvidenceObject, EvidencePackage, Sample, TestRecord	AIDCF-008	TR-09

Step	Name	Question answered	Primary AIDCF-014 classes	Detail owner	AIDCF-014 trace link
12	Control Assessment	What result does the evidence support, for which claim type and scope?	Assessment, AssessmentProcedure, ControlAssessment	AIDCF-009	TR-10
13	Finding / UNKNOWN Record	What deficiencies and unresolved uncertainties remain?	Finding, UnknownRecord, Action	AIDCF-008, AIDCF-009	TR-11
14	Gate Result	Are the non-compensating prerequisites met?	CriticalGate, GateResult	AIDCF-010	TR-12
15	Score Snapshot and Decision Recommendation	What do the results summarise to, and what does the engine recommend?	ScoreSnapshot, DecisionRecommendation	AIDCF-010	TR-13, TR-14
16	Assurance Decision	What has accountable authority decided, with which conditions and validity?	ReviewRecord, AssuranceDecision	AIDCF-010, AIDCF-011	TR-15
Loop	Report and Reassessment Trigger	How is the decision communicated, and what invalidates it?	AssuranceStatement, AssuranceReport, ReassessmentTrigger, Event	AIDCF-012, AIDCF-014	TR-16, TR-17, TR-18

CCM-REQ-006 Single chain. Artifacts shall use the sixteen step names exactly as listed above. Alternative names (for example "Critical Function", "Infrastructure Function", "Assessment Result", "Critical Gate" for step 14, "Score/Recommendation") shall be treated as aliases and replaced on revision.

CCM-REQ-007 Forward completeness. Each material element at steps 3 to 8 shall be linked forward to at least one element at the next step, or carry an UNKNOWN Record explaining why the link cannot be established.

CCM-REQ-008 Reverse traceability. Every Assurance Decision shall be reconstructable backwards to the Gate Results, Control Assessments, Evidence Objects and UNKNOWN Records on which it relies, using only recorded relationships from the AIDCF-014 registry.

9.1 Forward construction and reverse audit

The chain is built forwards (steps 1→16) and audited backwards (16→1). The forward pass is engineering analysis: it discovers what matters. The reverse pass is assurance review: it checks that every element of the decision is supported. The two passes find different defects. A forward pass that stops at step 9 produces a control list with no argument; a reverse pass that starts at a dashboard produces an argument with no discovery. AIDCF requires both.

9.2 Closing the loop

An Assurance Decision is valid only for the as-of date, scope and states it declares. Reassessment Triggers (step "Loop") are defined when the decision is issued and are evaluated against Events (changes, incidents, evidence expiry, new information). A fired trigger returns the case to step 2: the Capability Instance and its architecture are

re-baselined, the affected sub-claims are re-opened and the decision lifecycle moves to SUSPENDED, SUPERSEDED or EXPIRED as appropriate (Section 28.2).

10. Mapping to ISO/IEC/IEEE 15026-2 and GSN

ISO/IEC/IEEE 15026-2 specifies the minimum content and structure of an assurance case: claims, arguments that connect claims to evidence, the evidence itself, and the justifications, assumptions and context within which the argument holds. The Goal Structuring Notation (GSN), maintained as a community standard by the Assurance Case Working Group of the Safety-Critical Systems Club, provides a widely used graphical and structural notation for the same concepts. AIDCF adopts this structure. The mapping below is at concept level; it does not reproduce either standard's text.

Assurance-case concept	GSN element	AIDCF realisation	Chain step(s)	AIDCF-014 carrier
Top-level claim	Goal (top)	The top claim of Section 8	1–3	AssuranceDecision.subjectRef + scope; Assessment.decisionQuestion
Sub-claim	Goal	Outcome, function, dependency, scenario and safeguard claims	3–8	ServiceOutcome, RequiredFunction, Dependency, FailureScenario, Safeguard (each with a claim statement)
Argument / reasoning step	Strategy	Decomposition by Service Outcome × operating state, then by Required Function, then by Dependency and Fault Domain, then by Failure Scenario	between 3 and 8	Relationships HAS_OUTCOME, REQUIRES_FUNCTION, IMPLEMENTED_BY, DEPENDS_ON, SHARES_FAULT_DOMAIN_WITH, EXPOSED_TO, PROTECTED_BY
Requirement against which a claim is tested	(Goal at leaf level)	Control at a declared claim type	9	Control, GOVERNED_BY
Evidence	Solution	Evidence Requirement, Evidence Object, Control Assessment result	10–12	EvidenceRequirement, EvidenceObject, ControlAssessment; EVIDENCED_BY, SUPPORTS, ASSESSED_BY, RESULTS_IN
Context	Context	Assessment boundary, architecture baseline, declared operating states, exclusions, profile	2, 5, 6	Assessment.scope, ArchitectureSnapshot, OperatingState
Assumption	Assumption	Declared assumptions (for example utility fault level, design ambient conditions) recorded with owner and evidence status	any	Recorded on the relevant object; unsupported material assumptions become UNKNOWN Records
Justification	Justification	Rationale for a decomposition strategy, a sampling plan, a non-applicability or an alternative evidence route	any	ReviewRecord; Control.applicability; EvidenceRequirement.acceptableAlternatives
Defeater / challenge / counter-evidence	Dialectic elements (challenge, counter-claim)	CHALLENGES relationships from evidence; Findings; UNKNOWN Records	11–13	CHALLENGES, HAS_FINDING, HAS_UNKNOWN; Finding, UnknownRecord

Assurance-case concept	GSN element	AIDCF realisation	Chain step(s)	AIDCF-014 carrier
Undeveloped claim	Undeveloped goal	A sub-claim with no evidence path yet	any	UnknownRecord (type U1 or U5)
Away goal / module reference	Away goal	A claim supported in another case: AI Trust Graph, a provider's assurance, a separately assessed site	any	INTEGRATION_HANDOFF; IntegrationHandoff
Overall judgement of the case	(outside GSN core)	Gate Results, Score Snapshot, Decision Recommendation and Assurance Decision	14–16	GateResult, ScoreSnapshot, DecisionRecommendation, AssuranceDecision

Two AIDCF features have no direct GSN equivalent and are deliberate additions:

- **Critical Gates** are cross-cutting acceptance conditions over parts of the argument. In GSN terms each gate is a named sub-goal that must be supported regardless of how strongly sibling goals are supported; AIDCF makes that non-compensation explicit and machine-checkable.
- **UNKNOWN severity** gives every undeveloped or defeated element a defined effect on the top claim, so the case cannot be presented as complete while material elements are undeveloped.

> **CCM-REQ-009 Assurance-case form.** An AIDCF assessment shall be expressible as an assurance case in which the top claim is supported through sub-claims, strategies and evidence as mapped above. A tool may render the case in GSN or another notation, but the underlying records shall be the AIDCF-014 objects and relationships.
>
> **CCM-REQ-010 Defeaters are retained.** Challenging evidence, Findings and UNKNOWN Records shall be attached to the claim they defeat or weaken and shall not be deleted when resolved; resolution is recorded by CLOSED_BY, SUPERSEDES or a ReviewRecord.

11. How sub-claims decompose

The decomposition strategy is fixed so that two competent teams assessing the same Capability Instance produce arguments that can be compared.

Level A – Outcome × state. The top claim is decomposed into one sub-claim per Service Outcome per declared operating state: "Outcome Yi is achieved within Ti in state Sj." States that are declared but not supported by evidence are not dropped; they become undeveloped sub-claims with UNKNOWN Records.

Level B – Required Function. Each outcome sub-claim is decomposed into one sub-claim per Required Function: "Function Fk is delivered at the level outcome Yi requires in state Sj." For a training outcome typical functions are electrical supply to accelerator racks, heat removal from accelerators, scale-out fabric transport, checkpoint persistence, job scheduling, and control-plane availability. A **completeness sub-claim** sits beside them: "The set {Fk} is sufficient for Yi." It is supported by the functional decomposition record and architecture reconciliation and is the claim most directly protected by CG-01 and CG-02.

Level C – Dependency and Fault Domain. Each function sub-claim is decomposed by the dependencies of the systems implementing it: "Every material dependency of Fk in Sj is available at the required capacity, and no fault domain shared by redundant elements of Fk can defeat Fk beyond tolerance." Common-mode dependencies (shared source, controller, route, firmware, people) are argued here, not assumed away. A completeness sub-

claim "All material dependencies of Fk are identified" is supported by discovery coverage evidence and protected by CG-03.

Level D – Failure Scenario. Each dependency sub-claim is decomposed by credible Failure Scenarios drawn from AIDCF-004: "Under scenario Ψm, the residual state of Fk remains within tolerance." Scenarios include maintenance-induced, recovery-induced, compound and cross-domain paths.

Level E – Safeguard. Each scenario sub-claim is decomposed by the Safeguards positioned on its path: "Safeguard Gn prevents, detects, contains or recovers Ψm as required in Sj." The safeguard claim carries an implied claim type (a design claim for a pre-build decision, an effectiveness or resilience claim for an operational-resilience decision).

Leaf – Control at claim type. Each safeguard claim is tested through the AIDCF-007A Controls that govern the safeguard, each assessed at the claim type the Decision Question requires. The Control Assessment result is the leaf solution; its evidence is the Evidence Package.

Because a safeguard may sit on several scenario paths, and a control may govern several safeguards, the argument is a directed acyclic graph rather than a tree. AIDCF-014 relationships preserve the graph; a report may present a tree view per outcome.

```
flowchart TB
  TC["Top claim: X achieves Y within T in S for Q as of D"]
  CTX["Context: boundary, baseline, states, exclusions"]
  TC --- CTX
  TC --> A1["Outcome Y1 in NORMAL"]
  TC --> A2["Outcome Y1 in MAINTENANCE"]
  A2 --> B0["Completeness: functions sufficient"]
  A2 --> B1["Function: heat removal"]
  A2 --> B2["Function: electrical supply"]
  B1 --> C1["Dependency: CDU secondary loop, facility water, CDU controls"]
  B1 --> C0["Completeness: dependencies identified"]
  C1 --> D1["Scenario: CDU pump trip during maintenance"]
  D1 --> E1["Safeguard: N+1 pump with automatic changeover"]
  E1 --> L1["Control MCF-THM-010 at EFFECTIVENESS"]
  L1 --> EV["Evidence package: failover test, controller config, trend data"]
  EV -. challenged by .-> DF["Finding / UNKNOWN Record"]
```

Figure 1 – Illustrative decomposition (control identifier is illustrative of position; the authoritative control-to-safeguard mapping is in AIDCF-007A).

> **CCM-REQ-011 Decomposition order.** Assessments shall decompose the top claim in the order outcome × state, Required Function, Dependency and Fault Domain, Failure Scenario, Safeguard, Control. A different strategy may be used only with a recorded justification and shall still expose function, dependency and scenario sub-claims.
>
> **CCM-REQ-012 Completeness sub-claims.** Each outcome sub-claim shall carry a function-completeness sub-claim and each function sub-claim a dependency-completeness sub-claim, each supported by evidence of discovery coverage or by an UNKNOWN Record.
>
> **CCM-REQ-013 Undeclared states.** A sub-claim shall not be generalised to an operating state that was not declared and evidenced; declared but unevidenced states shall be carried as undeveloped sub-claims with UNKNOWN Records.

12. Assurance-case invariants

1. **One top claim per case.** Several Decision Questions about one Capability Instance are several cases that may share evidence.
2. **No orphan evidence.** Evidence that supports no claim is context, not support.
3. **No orphan controls.** A Control Assessment that cannot be traced to a safeguard claim may be reported, but it does not contribute to Gate Results through the argument; its gate contribution is limited to its declared AIDCF-007A gate mapping.
4. **Claims never exceed their evidence.** A claim's claim type, scope, states and period cannot be wider than those of the Control Assessments supporting it.
5. **Defeaters dominate until disposed.** An undisposed defeater (Finding, CHALLENGES evidence, US3+ UNKNOWN) on a decisive claim prevents that claim from being treated as supported.

> **CCM-REQ-014 Claim bounds.** No claim shall be stated with a wider scope, longer period, more operating states or a higher claim type than the evidence and Control Assessments supporting it.

Part IV – Core concepts

13. Conventions for concept definitions

Each concept below is given as: a **definition** (necessary and distinguishing meaning); **key attributes** (the minimum information that gives the concept assurance meaning; full property schemas are in AIDCF-003 Appendix C and AIDCF-014 object profiles); **invariants** (rules that must hold for every instance); and the **AIDCF-014 class** that carries it. Every instance additionally carries the AIDCF-014 common object envelope (identifier, type, semantic owner, version, status, provenance, effective period, classification, review state).

> **CCM-REQ-015 Concept fidelity.** Records representing a concept defined in this Part shall use the AIDCF-014 class named for it, shall carry the listed key attributes or an explicit UNKNOWN for each, and shall satisfy the listed invariants.

14. Decision and subject concepts

14.1 Decision Question

Definition. The specific decision that an assessment is commissioned to inform, stated with the decision owner, intended users and the consequence of the decision.

Key attributes. Question text; decision owner (role); intended users; decision type (for example design progression, commissioning acceptance, operational readiness, contract acceptance, expansion, post-incident reliance); required Assurance Level; required claim types; deadline or validity horizon.

Invariants. One Decision Question per assurance case. The Decision Question determines the required Assurance Level and claim types, never the other way round. A question that asks "is the facility safe/secure/resilient" without a capability, outcome and state bound is not a valid Decision Question (AIDCF-006 claim-language rules).

AIDCF-014 class. Carried as `Assessment.decisionQuestion` and on `AssuranceProgram`; repeated on `AssuranceDecision.question`.

14.2 Capability Instance

Definition. A specific, bounded deployment of an AI capability (training, inference, fine-tuning, HPC simulation, shared accelerator service) at defined locations, with a declared owner, architecture baseline, criticality and obligations.

Key attributes. capabilityInstanceId; name; capabilityType; owner; locations; lifecycle stage; Service Outcomes; criticality with rationale; assessment boundary (inclusions and explicit exclusions); operating envelope (load, environmental, capacity, duration, state constraints); architecture baseline reference; demand profile; effective period.

Invariants. Has at least one Service Outcome. Is an instance, not a type: "large-scale training" is a capability type; "TC-2 in Hall 3" is an instance. Every exclusion from its boundary identifies rationale, owner, residual dependency and effect on confidence (AIDCF-003 boundary model). Dependencies that cross the boundary remain represented.

AIDCF-014 class. `CapabilityInstance`.

14.3 Service Outcome

Definition. A measurable or assessable result that the Capability Instance is required to achieve, in one of the AIDCF-003 outcome families: availability, capacity, performance, integrity, isolation, recoverability, safety, efficiency, sovereignty, observability.

Key attributes. Outcome family; metric and unit; target; tolerance (Section 14.4); applicable operating states; measurement method and evidence source; owner; criticality.

Invariants. Every outcome has a tolerance. Outcomes are stated at capability level ("checkpoint written within the declared interval at the declared size"), not as component properties ("UPS is 2N"). A component property is an attribute of architecture, not an outcome.

AIDCF-014 class. `ServiceOutcome`; linked by `HAS_OUTCOME` from the Capability Instance.

14.4 Tolerance

Definition. The bounded deviation from a Service Outcome target that the decision owner accepts, expressed in the outcome's metric over a stated period and state, together with the maximum tolerable duration and frequency of breach.

Key attributes. Permitted degradation (magnitude); maximum duration; maximum frequency; applicable states; priority among outcomes when tolerance cannot be met for all (graceful-degradation order); approver.

Invariants. A tolerance is a decision-owner commitment, not an engineering estimate; engineering evidence demonstrates whether it can be met. Tolerances are capability-profile parameters and shall not be presented as universal values. A tolerance that is UNKNOWN makes the outcome sub-claim untestable (US5: it affects an outcome definition).

AIDCF-014 class. Attribute group of `ServiceOutcome` (and of CAP-03 Impact Tolerance control evidence).

15. Function and architecture concepts

15.1 Required Function

Definition. A function whose delivery at a stated level is necessary for a Service Outcome in a stated operating state, defined independently of the product or technology that implements it. Examples: deliver conditioned power to accelerator racks; remove heat from accelerator cold plates; transport collective-communication traffic between nodes; persist checkpoints durably; schedule and place jobs; provide control-plane authority over facility plant.

Key attributes. Function statement; required level (capacity, performance, integrity) per state; outcomes served; implementing systems; criticality; acceptable substitutes.

Invariants. Always called **Required Function** (not "Critical Function" or "Infrastructure Function"; those are legacy aliases). A Required Function is implemented by one or more systems; a system may implement several functions. Function before product: a function remains even when its implementing technology changes.

AIDCF-014 class. `RequiredFunction`; linked by `REQUIRES_FUNCTION` from the outcome and `IMPLEMENTED_BY` to systems.

15.2 Architecture

Definition. A versioned, as-of-date representation of the systems, assets, interfaces, resources and their relationships that implement the Required Functions of a Capability Instance, in a declared configuration perspective (designed, as-built, commissioned, observed, current approved, temporary or future; AIDCF-003 §48).

Key attributes. Capability reference; perspective; snapshot date; systems; assets; interfaces; resources; dependencies; fault domains; operating states; reconciliation status between perspectives.

Invariants. Designed, installed, commissioned and observed perspectives are distinct and are reconciled explicitly (AIDCF-006 CI-06). A claim about implementation, operation or effectiveness shall reference an as-built, commissioned or observed snapshot, not only the design.

AIDCF-014 class. `ArchitectureSnapshot`.

15.3 InfrastructureSystem

Definition. A coordinated set of assets and interfaces that together provide one or more Required Functions, for example a UPS system, a CDU and secondary loop, a parallel file system, a scale-out compute fabric, a BMS.

Key attributes. Functions implemented; member assets; subsystems; interfaces; redundancy groups; fault-domain memberships; owner and operator; capacity assertions.

Invariants. Classifying something as a system type does not establish its capacity, redundancy, condition or effectiveness (AIDCF-003 §29).

AIDCF-014 class. `InfrastructureSystem`; `COMPOSED_OF` assets.

15.4 Asset

Definition. An identifiable real-world item, service, person, procedure or resource relevant to a Required Function. The AIDCF-003 term is Asset Instance; "Asset" is used in prose.

Key attributes. assetId; assetClass; system membership; location; owner; operator; maintainer; lifecycle status; criticality (relative to a Capability Instance and outcome); capacity properties; interfaces; dependencies; fault

domains; evidence links; validity; confidence.

Invariants. Criticality is relative to a capability and outcome, never intrinsic to an asset class (AIDCF-003 IOT-REQ-028). People, procedures and contracts are assets when they are relied upon.

AIDCF-014 class. `AssetInstance`.

15.5 Interface

Definition. A defined point or mechanism of exchange between entities: electrical, thermal, hydraulic, network, mechanical, control, human/procedural or contractual/service (AIDCF-003 §31).

Key attributes. Interface class; endpoints; medium or protocol; capacity; direction; protection or isolation; authority (for control interfaces: signal, fail state, authentication).

Invariants. An interface enables exchange; a dependency is reliance. The two are never collapsed: a hydraulic interface between CDU and facility water does not, by itself, state whether the CDU depends on that water supply in every state.

AIDCF-014 class. `Interface`; `CONNECTS_TO`.

15.6 Resource

Definition. A consumable or constrained input required for operation: electrical energy, fuel, cooling capacity, water, coolant or refrigerant, space and structural capacity, network bandwidth, storage capacity, compute capacity, human capacity, time, spares (AIDCF-003 §32).

Key attributes. Resource type; quantity and unit; quality; source; replenishment; usable duration; constraints.

Invariants. Time is a resource: UPS autonomy, fuel autonomy, thermal ride-through and recovery time are resource assertions with units and states, not adjectives.

AIDCF-014 class. `Resource`; `CONSUMES_RESOURCE`.

15.7 Capacity Assertion

Definition. A statement of a capacity value for an entity, qualified by metric, unit, capacity term (nameplate, installed, commissioned, available, committed, resilient, transient, recovery; AIDCF-003 §40), operating state, environmental conditions, duration, derating assumptions, evidence date and confidence.

Invariants. Capacity terms are never interchangeable (IOT-REQ-042). A cross-domain capacity claim names its limiting dependency and any stranded capacity. A capacity assertion for a liquid-cooled rack row, for example, is bounded by the lowest of electrical distribution capacity, CDU heat-removal capacity at the declared facility-water supply temperature, and structural floor loading, and states which one limits.

AIDCF-014 class. `CapacityAssertion`.

16. Dependency, fault-domain and state concepts

16.1 Dependency

Definition. A directed reliance of one entity (the consumer) on another entity, resource, condition or decision (the provider), material in stated operating states.

Key attributes. Consumer; provider (or UNKNOWN); dependency class (electrical, thermal, hydraulic, mechanical/structural, network, control-system, timing/information, human/procedural, supplier/contractual, geographic/environmental, capacity, recovery-only); strength (absolute, material, conditional, supporting, recovery-only, UNKNOWN); applicable states; capacity requirement; substitutability; transfer mechanism and time; shared providers; owner; evidence; confidence; validity.

Invariants. Directional; not symmetric. A dependency crossing the assessment boundary remains represented as external or UNKNOWN; it does not disappear because it is out of scope (IOT-REQ-005). Contractual responsibility for a dependency is not evidence that it is reliable.

AIDCF-014 class. `Dependency`; `DEPENDS_ON`.

16.2 Fault Domain

Definition. A set of entities expected to be affected together by a common failure cause: physical (room, fire or flood zone, route), electrical (source, transformer, switchboard, UPS path, protection zone), thermal (air zone, coolant loop, CDU, chiller train), network (switch, plane, fabric, controller), storage (node group, metadata service, protection set), compute (node, tray, rack, scale-up domain, partition), control-system, software/configuration (firmware baseline, image, policy source), human/operational (team, shift, contractor) or external (utility, carrier, water, fuel, vendor, region).

Key attributes. Fault-domain type; member entities; common cause(s); containment boundary; evidence of isolation.

Invariants. Redundant elements that share a fault domain are not independent for failures of that domain. A firmware baseline shared by every accelerator node is a fault domain even though it has no physical extent.

AIDCF-014 class. `FaultDomain`; `SHARES_FAULT_DOMAIN_WITH`.

16.3 Operating State

Definition. A defined condition of operation of a Capability Instance, system or asset, from the canonical set NORMAL, DEGRADED, MAINTENANCE, EMERGENCY, RECOVERY, TRANSITION, SHUTDOWN, FAILED, UNKNOWN (Section 32).

Key attributes. State code; entity; entry trigger; authority; preconditions; required safeguards; expected duration; monitoring; exit criteria; evidence.

Invariants. Claims identify the states they cover and are not generalised to uncovered states (IOT-REQ-048). Temporary bypass and maintenance configurations are explicit states or configuration versions. Transitions are themselves assessable (a UPS transfer to static bypass is a TRANSITION).

AIDCF-014 class. `OperatingState`.

17. Adverse concepts

17.1 Threat

Definition. A deliberate actor, capability, intent or circumstance capable of adverse infrastructure consequence (AIDCF-004): for example, insider tampering with an EPMS breaker command, theft of accelerator trays in transit, a compromised remote-support session.

Invariants. Distinct from Hazard: deliberate sources require adversary assumptions (capability, access, intent) that non-deliberate sources do not.

AIDCF-014 class. `Threat`; `EXPOSED_TO`.

17.2 Hazard

Definition. A non-deliberate source, condition or event with potential for injury, damage, disruption or environmental impact: for example, arc flash at switchgear, lithium-ion battery thermal runaway, coolant leak onto energised equipment, refrigerant release, extreme ambient wet-bulb temperature, flood.

Invariants. Life-safety hazards are within CG-04 regardless of their effect on service outcomes.

AIDCF-014 class. `Hazard`; `EXPOSED_TO`.

17.3 Failure Mode

Definition. The manner in which a Required Function, asset, safeguard or dependency is lost, degraded, delayed, incorrect, intermittent or uncontrolled (AIDCF-004).

Invariants. Cause, mode, effect and consequence remain distinct. "UPS failed to transfer" is a mode; "static-switch firmware defect" is a cause; "rack PDUs de-energised" is a local effect; "training job lost to last checkpoint" is a consequence.

AIDCF-014 class. `FailureMode`.

17.4 Failure Scenario

Definition. An ordered, state-qualified sequence from an initiating threat, hazard or cause, through failure modes and propagation steps across dependencies and fault domains, past the safeguards that act or fail, to a residual state and its consequence for Service Outcomes.

Key attributes. Pre-state; initiating source; predisposing conditions; ordered steps; fault domains traversed; safeguards and their dependencies; residual state; outcome effects; recovery path; evidence; confidence; linked gates; reassessment triggers.

Invariants. Every material scenario identifies a Capability Instance and Service Outcome (THF-REQ-001). Unverified path elements remain visible as UNKNOWN. Path types (cascading, common-cause, maintenance-induced, recovery-induced, cross-domain, slow-burn) are recorded, because they drive which safeguards and states must be evidenced.

AIDCF-014 class. `FailureScenario`; `INITIATES`, `PROPAGATES_TO`, `RECOVERS_THROUGH`.

18. Safeguard and Control

These two concepts are the most frequently confused in assurance practice and are deliberately separated.

18.1 Safeguard

Definition. A real-world measure (physical, technical, procedural, organisational or contractual) intended to anticipate, prevent, detect, decide, respond, contain, withstand, degrade gracefully, recover or adapt with respect to one or more Failure Scenarios. Examples: an N+1 CDU pump with automatic changeover; a selective-

coordination protection scheme; a leak-detection loop with automatic valve isolation; a permit-to-work procedure for switching; a signed-firmware enforcement policy; an immutable checkpoint copy in a separate fault domain.

Key attributes. Safeguard function; protected subject; path position (cause, escalation, consequence or recovery side); activation; authority; dependencies (power, sensing, control, people, suppliers); states covered; capacity/duration; failure modes of the safeguard itself; evidence references; governing controls.

Invariants. Existence does not imply effectiveness (THF-REQ-005). A safeguard has its own dependencies and failure modes, which are analysed in the same way as any other element; a safeguard sharing a fault domain with what it protects is fragile.

AIDCF-014 class. `Safeguard`; `PROTECTED_BY` (protected entity PROTECTED_BY safeguard).

18.2 Control

Definition. A distinct, topic-specific normative requirement in the AIDCF-007A Canonical Master Control Registry, identified `MCF-<DOM>-NNN`, stating what an accountable role shall achieve with respect to a class of safeguards or assurance conditions, with independently testable assessment criteria and an evidence profile, assessed per claim type and per scope. Section 23 gives the full control model.

Distinction. The safeguard is the CDU pump changeover; the control is the requirement that CDU capacity and redundancy be demonstrated sufficient for the declared heat load in the declared states. A control can govern many safeguards; a safeguard can be governed by several controls. A control result says whether the safeguards it governs meet its criteria at the assessed claim type; it is not itself a safeguard.

AIDCF-014 class. `Control` within `ControlFamily` within `ControlDomain`; `GOVERNED_BY` (safeguard GOVERNED_BY control).

> **CCM-REQ-016 Safeguard–control separation.** Records shall not represent a safeguard and a control as the same object. Every Control Assessment that supports a safeguard claim shall reference the safeguard(s) assessed.

19. Evidence concepts

19.1 Evidence Requirement

Definition. A claim-specific statement of the evidence needed to support a Control Assessment at a claim type: required and optional evidence classes, quality floor, sufficiency floor, state and period coverage, independence, mandatory items, acceptable alternatives, UNKNOWN effect and reassessment triggers. The default is the control's AIDCF-007A evidence profile; a capability profile, gate or Decision Question may strengthen it, never weaken it.

AIDCF-014 class. `EvidenceRequirement`; `HAS_EVIDENCE_REQUIREMENT`.

19.2 Evidence Object

Definition. The smallest independently identified evidence item used to support, challenge or qualify an assertion (AIDCF-008): a protection-relay settings export, a load-bank test report, a seven-day CDU flow and differential-pressure trend, a fabric topology export, an interview record.

Key attributes. evidenceId; evidence class (E1–E7); source; collector; custodian; observation, collection and record times; scope; period; operating states covered; integrity level (I0–I5); quality vector (P F C R T I); restrictions; lifecycle state.

Invariants. An Evidence Object has no universal grade; quality and sufficiency are evaluated against a specific assertion (EVM-REQ-003). Derived evidence never exceeds the quality, scope and lineage of its sources.

AIDCF-014 class. `EvidenceObject`; `EVIDENCED_BY`, `SUPPORTS`, `CHALLENGES`, `QUALIFIES`.

19.3 Evidence Package

Definition. A governed collection of Evidence Objects addressing one control (at a claim type), function, system, scenario or gate, with a manifest of included, missing, substituted and restricted items, per-object assessments, package sufficiency (S0–S5), conflicts and UNKNOWNs, review and custody.

AIDCF-014 class. `EvidencePackage`.

19.4 Assertion

Definition. A precise, testable proposition (subject–predicate–object or equivalent) with claim type, scope and conditions, to which evidence is attached and on which a review records a state (CANDIDATE, SUPPORTED, CHALLENGED, QUALIFIED, CONFLICTING, UNKNOWN, SUPERSEDED; AIDCF-014 assertion vocabulary).

Invariants. Every sub-claim in the assurance case is an assertion. A management or vendor statement is an assertion with state CANDIDATE until evidence and review support it (AIDCF-006 CI-02).

AIDCF-014 class. No dedicated class in the AIDCF-014 v1.0 registry of 48 classes; carried as `assertionRefs` on EvidenceObject and as the target of SUPPORTS / CHALLENGES / QUALIFIES, with the AIDCF-014 assertion vocabulary. AIDCF-016 should introduce an explicit Assertion class.

20. Assessment-result concepts

20.1 Control Assessment

Definition. The recorded evaluation of one Control, at one claim type, for one declared scope (Capability Instance, boundary subset, states, period), producing exactly one control result (Section 26).

Key attributes. Control; claim type; scope; states; period; procedure and methods; samples; Evidence Package; result; criteria met/not met; Findings; UNKNOWN Records; assessor; reviewer; lifecycle state (for example SUSPENDED when reliance is withdrawn, with result reverting to UNKNOWN).

Invariants. One result per (control, claim type, scope). Results for different claim types of the same control are separate records. A CONFORMING result requires evidence at or above the evidence-requirement floor; otherwise the result is UNKNOWN. Implementation Maturity (IM0–IM5) may be recorded alongside but is never the result.

AIDCF-014 class. `ControlAssessment` (within an `Assessment`, using `AssessmentProcedure`, `Sample`, `TestRecord`); `ASSESSED_BY`, `RESULTS_IN`.

20.2 Finding

Definition. A recorded condition, supported by evidence, in which an assessment criterion, gate criterion or tolerance is not met or is at risk: criterion reference, condition, cause, consequence (to outcomes), classification,

gate effect, score effect, owner, status, and required action and retest.

Invariants. A Finding changes the underlying control result or gate evaluation; it is not a second, undeclared score deduction (AIDCF-010 no-double-counting). Accepted risk does not convert a nonconformity into conformity. A Finding is closed only by evidence and, where material, reassessment (`CLOSED_BY`).

AIDCF-014 class. `Finding`; `HAS_FINDING`, `REQUIRES_ACTION`.

20.3 UNKNOWN Record

Definition. A first-class record that a material fact, relationship, state, capacity, owner, evidence status or conclusion cannot be determined, stating subject, missing attribute, UNKNOWN type (U1–U10), UNKNOWN severity (US1–US5), reason, potential consequence, evidence needed, owner for resolution, status, decision and gate effect, and trigger for re-evaluation.

Invariants. UNKNOWN is never converted into pass, zero, fail, not applicable or accepted risk (IOT-REQ-006, EVM-REQ-006). Severity is assigned by the top-down test of Section 31 and carries a fixed decision effect. An UNKNOWN Record is closed only by evidence or by a reviewed disposition that does not assume a favourable value.

AIDCF-014 class. `UnknownRecord`; `HAS_UNKNOWN`.

21. Gate, score and decision concepts

21.1 Critical Gate

Definition. One of fifteen named, non-compensating prerequisites (CG-01 to CG-15; Section 35) protecting a condition without which no favourable decision is defensible, whatever the average performance.

AIDCF-014 class. `CriticalGate`; controls relate to gates through their AIDCF-007A gate mappings with role CRITICAL or SUPPORTING.

21.2 Gate Result

Definition. The evaluated status of one Critical Gate for one assessment scope: PASS, FAIL, UNKNOWN, APPROVED_NA or NOT_ASSESSED, with criterion, applicability, evidence references, state coverage, findings, UNKNOWNs, decision effect and review.

Invariants. Computed by the roll-up rule (Section 35.3); there is no conditional gate result. PARTIAL never yields PASS.

AIDCF-014 class. `GateResult`; `EVALUATES_GATE`, `CONSTRAINS`.

21.3 Score Snapshot

Definition. An immutable record of score inputs (control results), profile and weights, computed family/domain/capability indicators, coverage, band (or "Band withheld"), and gate and UNKNOWN overlays, with lineage.

Invariants. Scores never override gates and never create assurance. A snapshot is immutable; recalculation creates a new snapshot that SUPERSEDES the prior one.

AIDCF-014 class. `ScoreSnapshot`; `SUMMARISED_BY`.

21.4 Decision Recommendation

Definition. The engine- or assessor-produced proposed decision state, with the ceilings applied (gate, UNKNOWN, evidence, claim type) and the proposed conditions and limitations, before accountable review.

AIDCF-014 class. `DecisionRecommendation`; `RECOMMENDS`.

21.5 Assurance Decision

Definition. The bounded conclusion issued by an accountable decision authority, after review, answering the Decision Question with one decision state (REASONABLY_ASSURED, CONDITIONALLY_ASSURED, LIMITED, NOT_ASSURED, INCONCLUSIVE), conditions, limitations, ceilings, validity period, reassessment triggers and a decision lifecycle state.

Invariants. Never higher than the Decision Recommendation's ceiling; may be lower. Requires a named authority (`APPROVED_BY`) and review (`REVIEWED_BY`). "NOT ASSESSED" is not a decision state; where nothing was assessed, no decision is issued.

AIDCF-014 class. `AssuranceDecision` with `ReviewRecord`.

21.6 Assurance Statement and Assurance Report

Definition. The **Assurance Statement** is the short, controlled-wording communication of an Assurance Decision to named relying parties; the **Assurance Report** is the fuller record (one of the AIDCF-012 report classes) presenting the case, results, gates, scores, UNKNOWNs, limitations and actions.

Invariants. A report or statement shall not state or imply more than the decision it reports (AIDCF-014 BR-007).

AIDCF-014 class. `AssuranceStatement`, `AssuranceReport`; `REPORTED_IN`.

21.7 Reassessment Trigger

Definition. A declared condition that, when an Event satisfies it, invalidates or suspends reliance on identified claims, evidence or decisions and requires reassessment: material change, incident, failed test, evidence expiry, new dependency, supplier change, change in demand or density, new information.

AIDCF-014 class. `ReassessmentTrigger` with `Event`; `TRIGGERS_REASSESSMENT`, `INVALIDATED_BY`.

> **CCM-REQ-017 Decision authority.** No Assurance Decision shall be recorded without an approving accountable authority and a review record; a Decision Recommendation produced by a tool shall not be presented as an Assurance Decision.

Part V – Materiality and the control model

22. MATERIAL

> **Definition.** A fact, dependency, condition, deficiency or UNKNOWN is **material** if, had it been known or resolved differently, a competent assessor would reasonably expect it to change (a) a control result, (b) a

> gate result, (c) the decision state, or (d) a condition, limitation or ceiling attached to the decision, for the declared Capability Instance, Service Outcomes and operating states.

22.1 How to apply the definition

Materiality is a counterfactual test against the decision chain, not a judgement of size or cost. Apply it in four steps:

1. **Fix the reference frame.** Materiality is relative to the declared Capability Instance, Service Outcomes, tolerances and operating states. The same fact can be material to a training-readiness decision and immaterial to a colocation-capacity decision.
2. **Vary the item.** Consider the credible alternative values or resolutions of the item (it is true / false; the dependency exists / does not; the capacity is the asserted value / the lower bound of the evidence).
3. **Trace the effect.** Ask whether any alternative would change a result at steps 12, 14 or 16 of the chain, or a condition or ceiling on the decision.
4. **Record the answer.** If yes, the item is material and shall be represented (as a record, a Finding or an UNKNOWN Record). If no, it may be recorded as context but does not need decision treatment.

22.2 Worked tests (illustrative)

Item	Test	Material?
Generator fuel delivery contract names a single supplier whose depot is in the same flood zone as the site	A shared external fault domain could defeat the fuel-autonomy safeguard in a flood scenario; it could change the result of a generation-and-fuel control and the CG-05 evaluation.	Yes
A CRAH unit in an air-cooled hall that hosts no in-scope accelerators is overdue for filter replacement	No in-scope Required Function depends on it in any declared state.	No (context only)
Commissioning records for one of two CDU pumps cannot be located	Changes the evidence available for the CDU redundancy claim; could change a control result from CONFORMING to UNKNOWN and, if that control is CRITICAL for CG-06, the gate.	Yes
The out-of-band management network's console servers share an upstream switch with the in-band management network	Could defeat recovery of the control plane during an in-band failure; could change CG-10 and CG-15 evaluations.	Yes
Rack-PDU firmware version differs from the documented baseline but the vendor confirms no functional change, and the protection and metering behaviour has been tested on the installed version	The difference does not change any criterion outcome; it is a configuration-record Finding under change control only if CG-14 criteria require baseline currency.	Depends on CG-14 criteria; record and test against them

> **CCM-REQ-018 Materiality test.** Every decision to omit, exclude, defer or treat as context a fact, dependency, condition, deficiency or UNKNOWN shall be justified against the MATERIAL definition, with the reference frame stated.
>
> **CCM-REQ-019 No size proxy.** Materiality shall not be determined solely by cost, physical size, asset count or the number of affected records.

23. What a control is

23.1 The control model

A **control** in AIDCF is a distinct, topic-specific normative requirement, assessed per claim type and per scope. Concretely:

- **Distinct.** Each of the 360 controls addresses a different topic. MCF-POW-006 *Start Reliability* and MCF-POW-007 *Fuel Autonomy* are different requirements about different safeguards, not two depths of the same requirement.
- **Topic-specific.** The requirement names the accountable role and states what shall be achieved, in terms specific to its subject ("The facility owner shall…"), with 3–7 independently testable assessment criteria.
- **Normative.** It is a requirement (shall), not guidance; non-applicability requires approved rationale.
- **Assessed per claim type.** The same control can be assessed for DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS or RESILIENCE claims. Each assessment is a separate Control Assessment record with its own evidence and result.
- **Assessed per scope.** A control assessed for Hall 3 in NORMAL and MAINTENANCE states says nothing about Hall 4 or the RECOVERY state.

A control is not a lifecycle stage and not an assurance depth. Earlier registry drafts placed the four controls of each family at successive assurance depths (first control A2, second A3, third A4, fourth A5) so that, for example, "Governance Charter" could only ever be a design control and "Methodology Ownership" only an effectiveness control. That pattern is withdrawn. The claim type at which a control is assessed is selected by the Decision Question; the registry records, per control, the **required claim type** and **required Assurance Level** at which the control must be demonstrated to contribute positively to gates and decisions at that level.

23.2 Anatomy of a control record

The AIDCF-007A registry carries, for every control, the following fields (full schema in AIDCF-007A and AIDCF-014):

Field group	Content
Identity	controlId (`MCF-<DOM>-NNN`), familyId (`<DOM>-nn`), title
Requirement	objective; one or more specific "shall" statements naming the accountable role
Testability	assessmentCriteria (3–7, each independently testable); parameters (name, unit, illustrative default, basis); assessmentProcedure (ordered steps); assessmentMethods (Section 33.1)
Evidence	minimumEvidence; evidenceProfile (claim type, profile tier CP-1..CP-5, required and optional classes, quality floor, sufficiency floor, state and period coverage, independence, mandatory items, acceptable alternatives, UNKNOWN effect, reassessment triggers)
Classification	controlType (one primary type, Section 33.2); requiredAssuranceLevel (AL2–AL5, consistent with claim type); lifecycleStages; operatingStates
Applicability	when the control applies and when it does not
Gate linkage	gateMappings: list of {gateId, role CRITICAL or SUPPORTING}
Analysis linkage	failureModes addressed; ontologyClasses; relatedControls
Safety and UNKNOWN	testingSafety (authorisation needs for hazardous tests); unknownRule
References	informativeReferences (document-level only)

23.3 Control results and claim-type ordering

Claim types form an ordered ladder for each control, with one branch:

```
DESIGN → IMPLEMENTATION → OPERATION → EFFECTIVENESS
                                    ↘ RESILIENCE
```

A CONFORMING result at a claim type presupposes that each lower claim type on its path has been assessed CONFORMING for the same scope, in the same or an earlier still-valid assessment. If a lower claim type is PARTIAL, NONCONFORMING, UNKNOWN or NOT_ASSESSED, the higher claim type's result cannot be CONFORMING; it is recorded with the result its own evidence supports, capped as follows: a NONCONFORMING lower claim caps the higher at NONCONFORMING; PARTIAL caps at PARTIAL; UNKNOWN or NOT_ASSESSED caps at UNKNOWN. (Effectiveness demonstrated by test does not cure a design that does not meet its criteria; it shows that the deficient design happened to perform under the tested conditions.)

> **CCM-REQ-020 Control identity.** A control shall be a distinct topic-specific requirement. Registries shall not encode assurance depth or lifecycle stage through a control's position within its family.
>
> **CCM-REQ-021 Per-claim-type assessment.** Each Control Assessment shall record exactly one claim type and one scope. A result for one claim type shall not be reported as, or inferred to be, a result for another.
>
> **CCM-REQ-022 Claim-type presupposition.** A CONFORMING result at a claim type shall require CONFORMING results at each lower claim type on its path for the same scope; otherwise the cap in Section 23.3 applies.

23.4 Domains and families

The control universe is fixed at 15 domains, 90 families and 360 controls: six families per domain (`<DOM>-01` to `<DOM>-06`) and four controls per family, numbered `MCF-<DOM>-001` to `MCF-<DOM>-024`, where family *n* holds controls 4*n*−3 to 4*n*. Domain codes are fixed:

Code	Domain
GOV	Governance, Accountability and Assurance Direction
CAP	AI Capability, Service Outcome and Demand Management
DEP	Dependency, Concentration and External Reliance Management
PHY	Physical Site, Building and Security Infrastructure
POW	Electrical Power Infrastructure and Energy Continuity
THM	Cooling, Thermal and Fluid Infrastructure
ENV	Environmental and Resource Performance
CMP	AI Compute Infrastructure
STO	AI Storage and Data Infrastructure
NET	AI Network and Interconnect Architecture
OPS	Building Operations and Service Management
RES	Resilience, Continuity and Recovery
SUP	Supply Chain, OEM and Ecosystem Assurance
SEC	Digital, OT and Control-Plane Security

Code	Domain
EVA	Evidence, Assessment and Assurance Management

Domains organise assessment work and ownership; they do not partition the argument. A single Failure Scenario routinely traverses POW, THM, CMP, STO, NET, OPS and SEC.

Part VI – Controlled vocabularies

The tables in this Part are canonical. Their machine-readable encodings are the CSV files in `registries/vocabularies/`, which shall be kept identical in content to this Part. Legacy aliases are accepted on ingest only through the alias tables given here.

> **CCM-REQ-023 Vocabulary registry parity.** The CSV files in `registries/vocabularies/` shall encode exactly the codes, names and rules of Part VI; a change to either requires the same change to the other in the same release.

24. Claim types

Claim type	Meaning	Minimum Assurance Level
DESIGN	Requirement and design are adequate	AL2
IMPLEMENTATION	As-built/configured state reconciles to approved design	AL3
OPERATION	Operates as intended over a representative period and states	AL4
EFFECTIVENESS	Achieves its objective under tested/observed adverse conditions	AL5
RESILIENCE	Contains failure and supports recovery within tolerance	AL5

Legacy aliases. AIDCF-010 v1.0 "assurance dimensions" D1 Design, D2 Implementation, D3 Operation, D4 Effectiveness, D5 Resilience map 1:1 to the claim types above; the D-prefix shall not be used for claim types because it collides with the legacy domain codes D1–D8. AIDCF-008 §28 assertion claim types "intent/governance" map to DESIGN where the assertion concerns an approved requirement or design, and otherwise remain assertion types outside the control-result model ("absence", "compliance mapping", "decision").

25. Assurance Levels

Assurance Levels describe the depth of an assurance programme (AIDCF-011), and therefore the highest claim type a programme can support. They are not certification grades.

Level	Name	Permitted conclusion
AL0	Discovery	Inventory and decision discovery; no assurance conclusion
AL1	Readiness	Scope, ownership, applicability and evidence readiness; readiness conclusion only
AL2	Design	Requirements, architecture and safeguard design (DESIGN claims)
AL3	Implementation	Reconciliation to implemented/as-built state (up to IMPLEMENTATION claims)
AL4	Operational	Representative operation, monitoring, maintenance and incidents (up to OPERATION claims)

Level	Name	Permitted conclusion
AL5	Effectiveness and resilience	Safeguard effectiveness, failure containment and recovery (up to EFFECTIVENESS and RESILIENCE claims)
AL6	Continuing	Refresh of previously approved claims, evidence, gates and triggers; claims no higher than those already established at AL2–AL5

The registry field "Required Assurance Level" uses AL2–AL5 only.

Legacy code	Source	Canonical	Note
A2, A3, A4, A5	AIDCF-007A registry v1.0	AL2, AL3, AL4, AL5	1:1
A0 No assurance	AIDCF-006 §6	AL0 (programme) / INCONCLUSIVE (decision)	AIDCF-006 levels mixed programme depth and conclusion; use AL for depth and decision state for conclusion
A1 Descriptive	AIDCF-006 §6	AL0 or AL1	Descriptive inventory is AL0; readiness is AL1
A2 Design … A4 Operating	AIDCF-006 §6	AL2 … AL4	1:1
A5 Resilience	AIDCF-006 §6	AL5	AL5 covers effectiveness and resilience

26. Control result

Per control, per claim type, per scope.

Code	Meaning	Score value	Coverage treatment
CONFORMING	All assessment criteria met for the claim type, evidence ≥ profile floor	100	Scored
PARTIAL	Some criteria met; specific deficiencies recorded as Findings	50	Scored
NONCONFORMING	Criteria materially not met, or a safeguard demonstrated ineffective	0	Scored
UNKNOWN	Result cannot be determined (evidence below floor, conflict, inaccessible, etc.)	No value; excluded from mean	Counted in coverage denominator and in the UNKNOWN register
NOT_APPLICABLE	Approved non-applicability with rationale and approver	Excluded	Excluded from denominator
NOT_ASSESSED	In scope but not yet assessed	Excluded	Counted as coverage gap

Legacy aliases.

Legacy code	Canonical	Condition
NOT_IMPLEMENTED / "Not implemented"	NONCONFORMING	Always
INEFFECTIVE	NONCONFORMING	Always
IMPLEMENTED	CONFORMING	Valid only for DESIGN or IMPLEMENTATION claims; for higher claim types the legacy value maps to UNKNOWN pending reassessment
OPERATING_EFFECTIVELY / "Operating effectively"	CONFORMING	For OPERATION or EFFECTIVENESS claims

Legacy code	Canonical	Condition
STRONGLY_EFFECTIVE	CONFORMING	For OPERATION or EFFECTIVENESS claims
N/A, NA	NOT_APPLICABLE	Only where approved rationale and approver exist; otherwise NOT_ASSESSED
SUSPENDED	(lifecycle)	Set lifecycle state SUSPENDED on the ControlAssessment; result reverts to UNKNOWN
NOT ASSESSED, "Not Assessed"	NOT_ASSESSED	Always
PARTIAL (legacy score 25)	PARTIAL	Score value is now 50

Implementation Maturity (diagnostic only). IM0 Absent, IM1 Initial, IM2 Defined, IM3 Implemented, IM4 Operating, IM5 Optimised. IM levels describe how institutionalised a practice is (AIDCF-017). They are never a control result, never a score input and never a gate input.

> **CCM-REQ-024 Evidence floor governs result.** Where evidence for a Control Assessment is below the profile floor, the result shall be UNKNOWN; it shall not be recorded as PARTIAL, discounted or scored.

27. Gate result

Code	Meaning
PASS	All CRITICAL controls CONFORMING at required claim type, all gate criteria met with evidence ≥ floor, NONCONFORMING SUPPORTING controls evaluated with no defeat
FAIL	An in-scope CRITICAL control is NONCONFORMING or PARTIAL, or a gate criterion is demonstrated not met
UNKNOWN	No FAIL condition, but a CRITICAL control is UNKNOWN or NOT_ASSESSED, gate evidence is below floor, or a US4 UNKNOWN is attached
APPROVED_NA	Gate not applicable to the declared scope, with approved boundary rationale
NOT_ASSESSED	Applicable gate not yet evaluated

There is no conditional gate state. Legacy "APPROVED N/A" and "APPROVED_N/A" map to APPROVED_NA; legacy "CONDITIONAL" or "PASS WITH CONDITIONS" at gate level map to UNKNOWN or FAIL according to the roll-up rule, with any conditions moved to decision level.

28. Decision state and decision lifecycle

28.1 Decision state

Code	Meaning
REASONABLY_ASSURED	Sufficient appropriate evidence supports the bounded top claim; all applicable gates PASS or APPROVED_NA; no US3+ UNKNOWN undisposed
CONDITIONALLY_ASSURED	The bounded top claim is supported subject to named, owned, time-bound conditions; no gate FAIL or UNKNOWN
LIMITED	Some sub-claims supported; material limitations, a gate UNKNOWN or a gate FAIL whose AIDCF-010 ceiling is LIMITED prevent a broader conclusion
NOT_ASSURED	Evidence demonstrates that the top claim, or a controlling gate, is not met
INCONCLUSIVE	Evidence, method, boundary or definition cannot support any determination

28.2 Decision lifecycle

DRAFT → RECOMMENDED → APPROVED → ACTIVE, then one of SUSPENDED, WITHDRAWN, SUPERSEDED, EXPIRED.

Code	Meaning
DRAFT	Being prepared; no reliance
RECOMMENDED	Decision Recommendation issued for review
APPROVED	Approved by the accountable authority, not yet in effect
ACTIVE	In effect within its validity period
SUSPENDED	Reliance withdrawn pending reassessment (trigger fired, gate failed, evidence expired)
WITHDRAWN	Removed by the decision authority for integrity or governance reasons
SUPERSEDED	Replaced by a newer approved decision
EXPIRED	Validity period ended without renewal

Decision state and lifecycle are separate fields. A decision can be LIMITED and ACTIVE, or REASONABLY_ASSURED and SUSPENDED.

28.3 Aliases

Legacy	Canonical
ADVERSE; ADVERSE_NOT_ASSURED; "Adverse / Not assured"	NOT_ASSURED
UNKNOWN; UNKNOWN_INCONCLUSIVE; UNKNOWN_OR_INCONCLUSIVE; Withheld	INCONCLUSIVE
CONDITIONAL	CONDITIONALLY_ASSURED
"REASONABLY ASSURED" (with space)	REASONABLY_ASSURED
SUSPENDED, WITHDRAWN, SUPERSEDED (as decision states)	Decision lifecycle state of the same name; the decision state is retained
NOT ASSESSED	Not a decision state; no decision is issued

29. Confidence

Code	Name	Meaning
CONF-0	UNKNOWN	No defensible confidence state
CONF-1	LOW	Weak or limited evidence; high sensitivity to assumptions
CONF-2	MODERATE	Reasonable support with material limitations
CONF-3	HIGH	Strong support with no unresolved material conflict within scope
CONF-4	VERY_HIGH	Highly defensible, independently corroborated and representative support for the bounded claim
Flag CONFLICTING	—	Material evidence supports incompatible propositions
Flag STALE	—	Previously supported but no longer sufficiently current

Confidence is professional judgement with recorded rationale, not a computed probability (EVM-REQ-027) and never exceeds the ceiling set by the weakest mandatory evidence requirement or by an UNKNOWN (US2 caps at CONF-2).

Relationship to AIDCF-003 entity evidence status. AIDCF-003 §53 records an evidence status per entity or relationship (CONFIRMED, PARTIALLY_CONFIRMED, ASSERTED, INFERRED, CONFLICTING, STALE, UNKNOWN). That status is retained as a facet of records, and constrains confidence in claims relying on the record: ASSERTED or INFERRED (unreviewed) caps at CONF-1; PARTIALLY_CONFIRMED caps at CONF-2; CONFIRMED imposes no cap; CONFLICTING and STALE set the corresponding flags; UNKNOWN yields CONF-0. AIDCF-010 v1.0 confidence labels UNKNOWN, LOW, MODERATE, HIGH, VERY HIGH map to CONF-0 to CONF-4.

30. Evidence vocabularies

30.1 Evidence classes

Class	Name	Primary claim supported
E1	Documentary and declarative	Intent, governance, approved design
E2	Configuration and technical state	Implementation and technical state
E3	Operational and observational	Operation, performance, incidents over a period
E4	Validation and test	Effectiveness and resilience under stated conditions
E5	Independent and third-party	Independent corroboration
E6	Human and experiential	Practice, competence and context
E7	Derived and analytical	Analysis and relationship synthesis

30.2 Quality dimensions, bands, sufficiency and integrity

Each Evidence Object is rated 0–5 on six quality dimensions **for a specific assertion**: P provenance, F fitness/relevance, C coverage, R reliability, T timeliness/currentness, I integrity. The vector remains visible (for example `E4-P4-F4-C3-R4-T4-I3`).

Note: AIDCF-008 v1.0 labelled F as "Freshness" and T as "Traceability". The canonical assignment is F = fitness/relevance and T = timeliness/currentness. End-to-end traceability is enforced structurally by AIDCF-014 trace links TR-01 to TR-18 rather than rated per object.

Band	Name	Permitted use
Q0	Unusable	Context only; no positive assertion
Q1	Weak	Candidate/UNKNOWN; remediation priority
Q2	Limited	Descriptive or limited implementation claim
Q3	Adequate	Design/implementation claim where appropriate
Q4	Strong	Operating effectiveness with sufficient package coverage
Q5	Highly defensible	High-consequence gate or resilience claim, subject to method and sufficiency

Level	Name	Package supports
S0	None	No usable evidence
S1	Insufficient	Cannot support the claim

Level	Name	Package supports
S2	Minimum descriptive	Bounded description or design intent
S3	Reasonable	Design or implementation within declared scope
S4	Strong	Operating effectiveness for representative conditions
S5	Highly defensible	High-consequence gate/resilience claim with diverse, current, traceable, appropriately independent evidence

Level	Name	Anchor
I0	Unknown	Integrity method absent or unverified
I1	Basic controlled	Repository permissions and version history
I2	Verified copy	Source comparison or controlled export metadata
I3	Hashed	Cryptographic digest recorded in manifest
I4	Signed / trusted timestamp	Digital signature or trusted timestamp validated
I5	Tamper-evident custody	End-to-end signed custody, immutable storage or equivalent

30.3 Control profile tiers and evidence-system conformance

Tier	Name	Use	Minimum pattern
CP-1	Foundational	Governance, inventory, design intent	E1 plus accountable source; S2/S3
CP-2	Technical state	Configuration, topology, implementation	E1+E2 with observation/reconciliation; S3
CP-3	Operational	Operation, monitoring, maintenance	E2+E3 across representative period; S3/S4
CP-4	Effectiveness	Controls tested under defined conditions	E2+E3+E4; S4
CP-5	Critical/resilience	Critical gate, compound scenario, high-consequence outcome	Diverse E1–E4 and appropriate E5; S5 or profile-defined strong floor

Evidence-system conformance levels (how well an organisation's evidence system implements AIDCF-008; not a property of any single item of evidence) are **EC0–EC5**: EC0 Vocabulary aligned, EC1 Metadata complete, EC2 Provenance controlled, EC3 Assertion linked, EC4 Decision ready, EC5 Machine conformant. They were labelled E0–E5 in AIDCF-008 v1.0 and are renamed to avoid collision with evidence classes E1–E7.

The "evidence grade E0–E5" of AIDCF-005 §37 and AIDCF-006 §11 is replaced by quality bands Q0–Q5 (E0→Q0, E1→Q1, E2→Q2, E3→Q3, E4→Q4, E5→Q5, read as indicative; re-grade against the assertion where practicable).

31. UNKNOWN type and severity

31.1 Type (why it is unknown)

Code	Name	Definition
U1	Missing	Expected evidence not produced
U2	Inaccessible	Evidence exists but access is restricted or unavailable

Code	Name	Definition
U3	Stale	Evidence does not represent current state
U4	Conflicting	Material sources disagree
U5	Coverage gap	Required population, state, period or scenario not represented
U6	Source unverified	Origin or authority cannot be established
U7	Integrity uncertain	Alteration or custody cannot be ruled out
U8	Method uncertain	Collection or test method insufficiently defined
U9	Interpretation unresolved	Competent reviewers cannot yet determine meaning
U10	Dependency unknown	Evidence cannot establish a material external or shared dependency

31.2 Severity (what it does to the decision)

Apply the tests top-down; the first that applies sets the severity.

Code	Name	Test	Decision effect
US5	Systemic	Affects the boundary, a Capability or Outcome definition, or two or more gates	Decision ≤ INCONCLUSIVE
US4	Critical	Affects a gate-critical control or a gate criterion (CG-12 trigger)	Gate → UNKNOWN; decision ≤ LIMITED or INCONCLUSIVE per gate ceiling
US3	Material	Could change a control result or a decision condition, but no gate	Decision ≤ CONDITIONALLY_ASSURED; condition recorded
US2	Moderate	Affects confidence but not any result	Confidence ceiling CONF-2 for affected claims
US1	Low	No plausible effect on any result or condition	Record only

Type and severity are independent: a U3 (stale) UNKNOWN can be US1 or US5 depending on what it touches. Legacy AIDCF-010 v1.0 UNKNOWN materiality labels (Low, Moderate, Material, Critical, Systemic) map by name to US1–US5, with the canonical decision effects above replacing the v1.0 effects.

31.3 Worked severity tests (illustrative)

UNKNOWN	Type	Top-down test	Severity
The assessment boundary excludes the utility substation, but whether the site's two "diverse" feeds share a single upstream grid substation cannot be established	U10	Affects the boundary and two gates (CG-03 dependency visibility, CG-05 electrical supply)	US5
Integrated failover test records for the CDU pump changeover are missing; the relevant THM control is CRITICAL for CG-06	U1	Not boundary or definition; affects one gate-critical control	US4
Battery string impedance trend for one of four UPS modules is twelve months old; the UPS runtime control is SUPPORTING (not CRITICAL) for CG-05 and its result depends on that module	U3	No gate-critical control or gate criterion; could change a control result	US3
Interview evidence about shift handover conflicts with the handover log on wording, but both show handover occurred on every sampled shift	U4	Does not change any result; affects confidence in the operating-practice claim	US2
The model number of a non-critical environmental sensor in an office area inside the boundary is unrecorded	U1	No plausible effect on any result or condition	US1

> **CCM-REQ-025 UNKNOWN classification.** Every UNKNOWN Record shall carry one type (U1–U10) and one severity (US1–US5) assigned by the top-down test, and the decision effect of its severity shall be applied.

32. Operating states

Code	Definition
NORMAL	Approved architecture operates within declared envelope; no material planned impairment
DEGRADED	One or more functions, safeguards, paths or reserves are impaired while some capability remains
MAINTENANCE	A planned intervention changes availability, configuration, staffing or safeguards
EMERGENCY	Exceptional condition requiring urgent protective action
RECOVERY	Controlled restoration and validation after disruption
TRANSITION	Movement between architectures, states, locations or control regimes
SHUTDOWN	Intentional cessation with controlled residual dependencies
FAILED	Required function is not delivered
UNKNOWN	Current operating state cannot be determined

AIDCF-014 v1.0 omitted SHUTDOWN and FAILED; they are canonical.

33. Methods, control types, lifecycle stages, score bands

33.1 Assessment methods

Code	Meaning
INQUIRY	Questioning competent persons; corroboration required for technical claims
DOCUMENT_REVIEW	Examination of approved documents, drawings, studies and records
INSPECTION	Physical or logical examination of an asset, configuration or installation
OBSERVATION	Watching a process or activity being performed
RECONCILIATION	Comparing two representations (design vs as-built, register vs discovered state) and resolving differences
ANALYSIS	Engineering or data analysis (calculation, trend, model, dependency graph)
SAMPLING	Selecting a documented subset of a population for examination
TEST	Performing or witnessing a controlled test with defined initial state, method and expected result
REPERFORMANCE	Independently re-executing a procedure or calculation to confirm its result
SCENARIO_EXERCISE	Tabletop, simulation or live exercise of a defined scenario under authorisation

33.2 Control types

Code	Meaning
DIRECTIVE	Establishes required intent, authority, policy or criteria
PREVENTIVE	Stops an adverse event or condition from occurring

Code	Meaning
DETECTIVE	Identifies an adverse event or condition when it occurs
CORRECTIVE	Limits, contains or corrects the effect of an event or condition
RECOVERY	Restores the function or outcome after disruption

A control has one primary type.

33.3 Lifecycle stages

Code	AIDCF-003 §17 legacy stage(s)
PLAN	Concept
DESIGN	Design
BUILD	Construction
COMMISSION	Commissioning; Operational readiness (acceptance and handover)
OPERATE	Operation
CHANGE	Change
DECOMMISSION	Decommissioning

Lifecycle stage (where in the asset or capability life) is independent of claim type (what is being claimed) and of object lifecycle status (DRAFT, CANDIDATE, REVIEWED, APPROVED, ACTIVE, SUSPENDED, WITHDRAWN, SUPERSEDED, RETIRED in AIDCF-014).

33.4 Score bands

Band	Name	Indicator range
B5	Strong	≥ 90
B4	Substantial	75 – 89.99
B3	Moderate	50 – 74.99
B2	Weak	25 – 49.99
B1	Very weak	< 25
WITHHELD	Band withheld (coverage/UNKNOWN)	Coverage < 80% or a US4/US5 UNKNOWN in scope

AIDCF-010 v1.0 bands (B5 90–100 Strong, B4 80–89 Established, B3 65–79 Developing, B2 45–64 Weak, B1 0–44 Critical weakness, BN) are superseded; scores computed under the old bands are not comparable and start a new trend series.

34. Relationship vocabulary

The 41 relationship types of AIDCF-014 §9 are canonical: HAS_OUTCOME, REQUIRES_FUNCTION, IMPLEMENTED_BY, COMPOSED_OF, CONNECTS_TO, CONSUMES_RESOURCE, DEPENDS_ON, SHARES_FAULT_DOMAIN_WITH, LOCATED_IN, OPERATED_BY, CONTROLLED_BY, MONITORED_BY, PROTECTED_BY, EXPOSED_TO, INITIATES, PROPAGATES_TO, RECOVERS_THROUGH, GOVERNED_BY, HAS_EVIDENCE_REQUIREMENT, EVIDENCED_BY, SUPPORTS, CHALLENGES, QUALIFIES, ASSESSED_BY, RESULTS_IN, HAS_FINDING, HAS_UNKNOWN, EVALUATES_GATE, CONSTRAINS, SUMMARISED_BY, RECOMMENDS, REVIEWED_BY, APPROVED_BY, REPORTED_IN, REQUIRES_ACTION,

CLOSED_BY, INVALIDATED_BY, TRIGGERS_REASSESSMENT, SUPERSEDES, MIGRATED_FROM, INTEGRATION_HANDOFF.

Legacy	Canonical	Rule
SUPPORTS_OUTCOME	IMPLEMENTED_BY (inverse)	Function IMPLEMENTED_BY System; for outcome links use HAS_OUTCOME / REQUIRES_FUNCTION
IMPLEMENTS	IMPLEMENTED_BY (inverse)	Swap subject and object
CONSTRAINS_GATE; LIMITS_DECISION	CONSTRAINS	—
PROTECTS	PROTECTED_BY (inverse)	Swap subject and object
DISPUTES	CHALLENGES	—
EVALUATES	ASSESSED_BY (inverse)	ControlAssessment EVALUATES Control becomes Control ASSESSED_BY ControlAssessment
TESTED_BY	EVIDENCED_BY	Target is a TestRecord-derived Evidence Object
RECOVERED_BY	RECOVERS_THROUGH	—

AIDCF-003 finer-grained predicates (POWERED_BY, COOLED_BY, SUPPLIED_BY, AUTHENTICATED_BY, TIMED_BY and the common-mode predicates) remain valid as AIDCF-003 specialisations; on exchange they are carried as DEPENDS_ON or SHARES_FAULT_DOMAIN_WITH with the specific predicate recorded as the dependency class or a qualifier. Namespace: `https://w3id.org/aidcf/ontology#` (prefix `aidcf:`), version IRI `https://w3id.org/aidcf/ontology/1.3`. Registration of the w3id namespace is pending.

Part VII – Gate, decision and score doctrine (summary)

This Part states the doctrine that every artifact shall respect. Gate criteria, per-gate decision ceilings, evaluation procedures and score computation are specified in AIDCF-010; where AIDCF-010 adds detail, it does so within these rules.

35. Critical Gates

35.1 The fifteen gates

Gate	Name	Protects	Evidence floor
CG-01	Capability and outcome definition	Capability Instance, Service Outcomes, tolerances, criticality and Decision Question are defined, owned and approved	S3
CG-02	Boundary and architecture integrity	Assessment boundary, current architecture, as-built reconciliation, exclusions	S3
CG-03	Critical dependency visibility	Material internal and external dependencies, common-mode and concentration identified and owned	S4
CG-04	Life safety and hazardous energy	Fire, electrical arc flash and shock, hazardous fluids and chemicals, confined space, refrigerants, battery thermal runaway, safe work	S4
CG-05	Resilient electrical supply	Utility, distribution, UPS/BESS, generation, fuel, protection and power quality sufficient for the outcome across states	S5

Gate	Name	Protects	Evidence floor
CG-06	Thermal continuity	Heat-removal path (air, liquid, facility water, heat rejection) sufficient across states including transients	S5
CG-07	Critical control-system integrity	BMS, EPMS, CDU, PLC, SCADA and automation function safely, fail safe, are configured and governed	S4
CG-08	Compute, storage and network functional integrity	Accelerators, storage and fabrics deliver required function, performance and integrity	S4
CG-09	Failure containment	Fault domains, isolation and blast-radius limits prevent propagation beyond tolerance	S4
CG-10	Recovery readiness	Restoration of the outcome (power, cooling, compute, data/checkpoint, network, control plane) within tolerance is planned and tested	S5
CG-11	Evidence sufficiency	Evidence for decisive claims meets floors	S4
CG-12	Critical UNKNOWN	No unresolved US4/US5 UNKNOWN without disposition	S3
CG-13	Responsibility and authority	Every critical function and dependency has an accountable owner and decision authority	S3
CG-14	Change control	Material changes are controlled, impact-assessed and validated	S4
CG-15	Integrated digital authority over infrastructure	Identity, privilege, remote access and control-plane reach over facility and AI infrastructure is bounded, monitored and recoverable	S4

Gate evidence profiles (classes and mandatory items per gate) are in AIDCF-008 Part XIII.

35.2 Control-to-gate mapping

Each control lists its gate mappings individually, each with a role:

- **CRITICAL** – the gate cannot PASS unless this control is CONFORMING at its required claim type, when in scope. Used only where failure of the control directly defeats the gate's protected condition.
- **SUPPORTING** – the control informs the gate evaluation; a NONCONFORMING result requires a documented gate-level evaluation but does not automatically fail the gate.

A control may map to zero or more gates. The meta-gates CG-11 and CG-12 are evaluated across all controls and are not mapped per control. Every other gate has at least three CRITICAL controls across the registry, mapped by meaning across domains (for example, CG-04 draws on POW, THM, PHY and battery or refrigerant controls in ENV and STO; CG-10 on POW, THM, CMP, STO, NET and RES recovery controls). Whole-domain mapping, in which every control in a domain was mapped to the same gates, is withdrawn.

35.3 Gate roll-up rule

For a gate in scope, evaluate in order:

1. **FAIL** if any in-scope CRITICAL control is NONCONFORMING or PARTIAL, or a gate criterion is demonstrated not met.
2. Otherwise **UNKNOWN** if any in-scope CRITICAL control is UNKNOWN or NOT_ASSESSED, or gate evidence is below the floor, or a US4 UNKNOWN is attached to the gate.
3. Otherwise **PASS** if all CRITICAL controls are CONFORMING at the required claim type, all gate criteria are met with evidence at or above the floor, and every SUPPORTING control that is NONCONFORMING has a

documented gate-level evaluation concluding no defeat.
4. **APPROVED_NA** only with approved boundary rationale.

PARTIAL never yields PASS. There is no conditional gate result.

> **CCM-REQ-026 Gate non-compensation.** No score, weighting, maturity level, accepted risk or volume of conforming results elsewhere shall change a Gate Result.
>
> **CCM-REQ-027 Gate roll-up.** Gate Results shall be computed by the roll-up rule of Section 35.3; any criterion-level judgement shall be recorded with evidence and reviewer.

36. From results to decision

The Decision Recommendation is the most favourable decision state that no applicable ceiling forbids. Ceilings come from four sources, and the lowest applies:

Source	Ceiling
Gate FAIL	No higher than LIMITED; AIDCF-010 assigns per gate whether FAIL yields LIMITED or NOT_ASSURED
Gate UNKNOWN (including via US4)	No higher than LIMITED, or INCONCLUSIVE where the AIDCF-010 gate ceiling so specifies
Applicable gate NOT_ASSESSED	The gate's AIDCF-010 UNKNOWN ceiling, as for Gate UNKNOWN; NOT_ASSESSED is never treated as PASS or APPROVED_NA, so no favourable decision on the top claim
UNKNOWN severity	US5 → INCONCLUSIVE; US4 → per gate; US3 → CONDITIONALLY_ASSURED with condition recorded; US2 → confidence ≤ CONF-2
Claim type and Assurance Level	The decision cannot assert a claim type above the programme's Assurance Level or above the claim types assessed
Top-claim completeness	Missing top-claim element (Section 8) → INCONCLUSIVE

CONDITIONALLY_ASSURED requires that every condition be specific, owned, time-bound and verifiable, and that no gate is FAIL or UNKNOWN. A decision authority may issue a lower state than recommended, never a higher one; departures are recorded with rationale.

> **CCM-REQ-028 Ceiling discipline.** An Assurance Decision shall not exceed the lowest applicable ceiling; overrides upward are prohibited, and overrides downward shall record rationale and authority.
>
> **CCM-REQ-029 Conditions.** Every condition attached to a CONDITIONALLY_ASSURED decision shall name an owner, a completion date or trigger, and the evidence that will demonstrate it is met.

37. Score doctrine

1. **Control score** = the control result's score value (CONFORMING 100, PARTIAL 50, NONCONFORMING 0). UNKNOWN, NOT_APPLICABLE and NOT_ASSESSED have no value.
2. **Family and domain score** = mean of scored controls, unweighted unless an approved capability profile sets weights before results are known.
3. **Coverage** = scored ÷ (in-scope − NOT_APPLICABLE), reported with every score. UNKNOWN and NOT_ASSESSED controls are in the denominator.

4. **Bands** (Section 33.4) are reported only if coverage ≥ 80% and there is no US4/US5 UNKNOWN in that scope; otherwise "Band withheld (coverage/UNKNOWN)".
5. **Evidence does not multiply the score.** Insufficient evidence makes the result UNKNOWN. The AIDCF-010 v1.0 evidence-confidence factor (S-factor multiplier) is withdrawn.
6. **Scores never override gates** and never raise a decision; they summarise, prioritise and trend.
7. **Comparability** requires the same profile version, scope, claim types and states; a change to any starts a new series.

> **CCM-REQ-030 Score presentation.** A score shall not be published without its coverage, its band or "Band withheld" status, the applicable Gate Results and the count of UNKNOWN Records by severity for the same scope.

Part VIII – Ladders crosswalk

38. Every level scale used in AIDCF

AIDCF artifacts define many ordinal scales. They measure different things, and several share letters. The table below lists every scale, its owner, what it is about and whether it is normative for assurance decisions.

Key to "Decision role": *Decisive* – directly determines results, gates or decision state. *Constraining* – sets floors or ceilings on decisive values. *Descriptive* – informs planning or presentation; never changes a result. *Conformance* – describes how well an artifact, dataset, tool or organisation implements AIDCF; says nothing about the facility.

Scale	Values	What it measures	Owner	Decision role	Relationship / notes
Control result	CONFORMING, PARTIAL, NONCONFORMING, UNKNOWN, NOT_APPLICABLE, NOT_ASSESSED	Outcome of one Control Assessment	AIDCF-002 / AIDCF-010	Decisive	Input to gates and scores
Gate result	PASS, FAIL, UNKNOWN, APPROVED_NA, NOT_ASSESSED	Status of one Critical Gate	AIDCF-002 / AIDCF-010	Decisive	Sets decision ceilings
Decision state	REASONABLY_ASSURED … INCONCLUSIVE	Answer to the Decision Question	AIDCF-002 / AIDCF-010	Decisive	Bounded by ceilings
Decision lifecycle	DRAFT … EXPIRED	Status of a decision record	AIDCF-002 / AIDCF-014	Constraining	SUSPENDED/EXPIRED remove reliance
Claim type	DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS, RESILIENCE	What a Control Assessment asserts	AIDCF-002	Decisive	Ordered (Section 23.3); minimum AL per type
Assurance Level	AL0–AL6	Programme depth	AIDCF-011	Constraining	Caps the highest claim type a programme can support

Scale	Values	What it measures	Owner	Decision role	Relationship / notes
Legacy APS assurance level	A0–A5	Mixed depth/conclusion	AIDCF-006 v1.0	Superseded	Map to AL (Section 25)
Legacy assurance dimension	D1–D5	Claim depth	AIDCF-010 v1.0	Superseded	Map to claim types (Section 24)
UNKNOWN severity	US1–US5	Decision effect of an UNKNOWN	AIDCF-002	Decisive	Top-down test (Section 31.2)
UNKNOWN type	U1–U10	Reason for an UNKNOWN	AIDCF-008	Descriptive	Drives resolution action, not effect
Confidence	CONF-0–CONF-4 + CONFLICTING, STALE	Reviewer confidence in a claim	AIDCF-008	Constraining	REASONABLY_ASSURED expects CONF-3 or higher on decisive claims (AIDCF-010)
Entity evidence status	CONFIRMED … UNKNOWN	Evidence status of a record	AIDCF-003	Constraining	Caps confidence (Section 29)
Evidence class	E1–E7	Kind of evidence	AIDCF-008	Constraining	Required classes per profile
Quality dimension ratings	P, F, C, R, T, I each 0–5	Fitness of one object for one assertion	AIDCF-008	Constraining	Quality floor per profile
Coverage rating	C0–C5	Anchor scale for the C dimension	AIDCF-008 §24	Constraining	Same scale as dimension C
Quality band	Q0–Q5	Overall usability of one object for one assertion	AIDCF-008	Constraining	Replaces evidence grade E0–E5
Legacy evidence grade	E0–E5	Evidence strength	AIDCF-005 §37, AIDCF-006 §11	Superseded	Map to Q0–Q5
Sufficiency	S0–S5	Whether a package supports a claim	AIDCF-008	Constraining	Floors per control profile and per gate
Integrity	I0–I5	Protection against alteration	AIDCF-008	Constraining	Feeds dimension I
Control profile tier	CP-1–CP-5	Evidence pattern of a control profile	AIDCF-008	Constraining	Sets default classes and S floor
Evidence-system conformance	EC0–EC5	Maturity of an evidence system's implementation of AIDCF-008	AIDCF-008	Conformance	Formerly E0–E5 in AIDCF-008 §4
Score band	B1–B5, Band withheld	Presentation of a score	AIDCF-010	Descriptive	Never changes a result or gate
Implementation Maturity	IM0–IM5	Institutionalisation of a practice	AIDCF-017	Descriptive	Diagnostic only
Consequence severity	S0–S5 (Negligible … Catastrophic)	Severity of scenario consequence	AIDCF-004 Part XI	Constraining (scenario prioritisation)	Letter collides with sufficiency; cite as THF-S0–THF-S5

Scale	Values	What it measures	Owner	Decision role	Relationship / notes
Safe validation hierarchy	V0–V7	Intrusiveness of a validation method	AIDCF-004	Descriptive (method selection)	Higher is more intrusive, not "better"; selection governed by safety and authority
Ontology conformance	C0–C4	Implementation of AIDCF-003	AIDCF-003	Conformance	Collides with coverage C0–C5; cite as IOT-C0–IOT-C4
Threat-model conformance	F0–F4	Implementation of AIDCF-004	AIDCF-004	Conformance	Collides with dimension F; cite as THF-F0–THF-F4
Integration conformance	I0–I5	Implementation of AIDCF-005	AIDCF-005	Conformance	Collides with integrity I0–I5; cite as INT-I0–INT-I5
Principles conformance	P0–P5	Implementation of AIDCF-006	AIDCF-006	Conformance	Collides with dimension P and AIDCF-011 P0–P9; cite as APS-P0–APS-P5
Traceability conformance	T0–T7	Implementation of AIDCF-014	AIDCF-014	Conformance	Collides with dimension T; cite as TDS-T0–TDS-T7
Architecture conformance	AR0–AR6	Implementation of AIDCF-013	AIDCF-013	Conformance	—
Architecture layers	L0–L9	Reference-architecture layer	AIDCF-013	Descriptive	Not a level; a classification
Programme lifecycle	P0–P9	Stage of an assurance programme	AIDCF-011	Descriptive	Cite as APF-P0–APF-P9
Programme stage gates	PG-0–PG-9	Programme progression checkpoints	AIDCF-011	Constraining (programme)	Not Critical Gates
Assessor competence	AC-0–AC-6	Role competence level	AIDCF-009 v1.1	Constraining (who may sign)	—
Report class	R1–R10	Report type	AIDCF-012	Descriptive	A class, not a level
Canonical queries	Q01–Q12	Graph query families	AIDCF-014	Descriptive	Two-digit codes; not quality bands
Critical Gate identifiers	CG-01–CG-15	Gate identity	AIDCF-002 / AIDCF-010	—	Identifiers, not a scale

39. Which ladders govern decisions

1. **Normative for the decision:** control result (per claim type), Gate Result, UNKNOWN severity and decision state. These four, combined by the rules of Part VII, determine the Assurance Decision.
2. **Normative as constraints:** Assurance Level and claim type (what may be claimed); sufficiency, quality floors, evidence classes and CP tiers (when a result can be other than UNKNOWN); confidence (whether REASONABLY_ASSURED is supportable); AC levels (who may sign).
3. **Informative only:** score bands, IM levels, AIDCF-004 V-levels, report classes, architecture layers.
4. **Not about the facility at all:** every artifact conformance scale (IOT-C, THF-F, INT-I, APS-P, EC, TDS-T, AR). A dataset can be TDS-T5 conformant and describe a facility whose decision is NOT_ASSURED.

> **CCM-REQ-031 Ladder qualification.** Where a level code could be confused with another scale (C, F, I, P, S, T, E, Q, D prefixes), it shall be written with its owning artifact's short code (for example THF-S3, IOT-C2, INT-I3, APS-P2, TDS-T5) or with the scale named in full.
>
> **CCM-REQ-032 No ladder substitution.** No conformance, maturity, score or validation-method level shall be presented as, converted into, or used in place of a control result, Gate Result or decision state.

Part IX – Artifact register and precedence graph

40. Artifact identifier register

Public ID (`AIDCF-NNN`) is used in filenames and cross-references. The short code is used only to form requirement IDs (`<SHORT>-REQ-NNN`).

Public ID	Title	Short code	Location
AIDCF-001	AIDCF Manifesto	MAN	foundations/manifesto/
AIDCF-002	Core Conceptual Model	CCM	foundations/conceptual-model/
AIDCF-003	Infrastructure Ontology and Taxonomy	IOT	foundations/ontology-taxonomy/
AIDCF-004	Threat, Hazard and Failure Model	THF	foundations/threat-model/
AIDCF-005	AIDCF–AI Trust Graph Integration Standard	INT	foundations/aitg-integration/
AIDCF-006	Assurance Principles Standard	APS	foundations/assurance-principles/
AIDCF-007A	Control Universe Constitutional Amendment and Canonical Master Control Registry	CTL	standards/aidcf-007a/
AIDCF-008	Evidence Model	EVM	standards/aidcf-008-evidence-model/
AIDCF-009	Assessment Methodology Practitioner Handbook (v1.0 core and v1.1 enhanced companion)	AMH	standards/aidcf-009-assessment-methodology/
AIDCF-010	Scoring and Decision Framework	SDF	standards/aidcf-010-scoring-framework/
AIDCF-011	Assurance Program Framework	APF	standards/aidcf-011-assurance-program/
AIDCF-012	Reporting and Assurance Statement Standard	RAS	standards/aidcf-012-reporting-standard/
AIDCF-013	AI Data Centre Assurance Reference Architecture	ARA	standards/aidcf-013-reference-architecture/
AIDCF-014	End-to-End Traceability and Machine-Executable Data Standard	TDS	standards/aidcf-014-traceability-standard/
AIDCF-015	Reference Implementation, Synthetic Dataset and Conformance Pilot	RIM	implementation/
AIDCF-016	Machine-Executable Data Model Standard	MDM	standards/aidcf-016-machine-executable-model/
AIDCF-017	Capability Assurance Maturity Model	CMM	standards/aidcf-017-maturity-model/
AIDCF-018	Domain Guides (one per domain, AIDCF-018-)	DG	standards/aidcf-018-domain-guides/

Public ID	Title	Short code	Location
AIDCF-019	Standards Crosswalks	XW	crosswalks/
AIDCF-VR-001	Full-Stack Validation Report (formerly mislabelled AIDCF-016)	VR	validation/validation-report/
AIDCF-WP-001	Technical Whitepaper	WP	whitepaper/current/

"AIDCF-007" without suffix refers only to the superseded v1.0 control catalogue. All live references use AIDCF-007A.

41. Legacy identifiers

Legacy identifier	Canonical public ID
AIDCF-MAN-001	AIDCF-001
AIDCF-IOT-001	AIDCF-003
AIDCF-THF-001	AIDCF-004
AIDCF-INT-001	AIDCF-005
AIDCF-APS-001	AIDCF-006
AIDCF-MCF-001; "AIDCF-007" (as a live reference)	AIDCF-007A
AIDCF-EVM-001	AIDCF-008
AIDCF-AM-001	AIDCF-009
AIDCF-SDF-001	AIDCF-010
AIDCF-APF-001	AIDCF-011
AIDCF-RAS-001	AIDCF-012
AIDCF-ARA-001	AIDCF-013
AIDCF-TDS-001	AIDCF-014
AIDCF-RIP-001	AIDCF-015
"AIDCF-016" applied to the Full-Stack Validation Report	AIDCF-VR-001
"Artifact 003" … "Artifact 016" ordinal references	The corresponding AIDCF-NNN

42. Dependency and precedence graph

Arrows point from an artifact to the artifacts that depend on it for meaning. Precedence follows the arrows downwards.

```
flowchart TB
  A001[AIDCF-001 Manifesto] --> A002[AIDCF-002 Core Conceptual Model]
  A002 --> A003[AIDCF-003 Ontology and Taxonomy]
  A003 --> A004[AIDCF-004 Threat, Hazard and Failure Model]
  A004 --> A005[AIDCF-005 AITG Integration]
  A005 --> A006[AIDCF-006 Assurance Principles]
  A006 --> A007A[AIDCF-007A Control Registry]
  A006 --> A008[AIDCF-008 Evidence Model]
  A007A --> A008
  A008 --> A009[AIDCF-009 Assessment Methodology]
  A007A --> A009
  A009 --> A010[AIDCF-010 Scoring and Decision]
  A010 --> A011[AIDCF-011 Assurance Program]
  A011 --> A012[AIDCF-012 Reporting]
  A003 --> A013[AIDCF-013 Reference Architecture]
  A007A --> A013
  A002 --> A014[AIDCF-014 Traceability and Data]
  A014 --> A016[AIDCF-016 Machine-Executable Model]
  A014 --> A015[AIDCF-015 Reference Implementation]
  A016 --> A015
  A011 --> A017[AIDCF-017 Maturity Model]
  A007A --> A018[AIDCF-018 Domain Guides]
  A007A --> A019[AIDCF-019 Crosswalks]
  A015 --> VR[AIDCF-VR-001 Validation Report]
  A010 --> WP[AIDCF-WP-001 Whitepaper]
  A002 -. vocabularies .-> REG[(registries/vocabularies)]
```

AIDCF-002 has a direct semantic link to every downstream artifact (omitted from the figure for legibility) through the controlled vocabularies.

Part X – Legacy eight-domain to fifteen-domain mapping

43. Purpose and status

AIDCF-001 Part V and AIDCF-003/AIDCF-004 v1.0 organised the framework into eight domains (D1–D8). The eight-domain model is superseded by the fifteen domains of Section 23.4. This mapping is retained only so that legacy material can be traced; new material shall use the fifteen domains.

The eight legacy domains split facility engineering, power and cooling in ways that hid their dependencies, and left governance, capability definition, dependency management, resilience, supply chain, digital and OT security, and evidence management as cross-cutting themes without owners. The fifteen-domain model gives each of those an owning domain.

44. Mapping table

Legacy domain	Legacy concern (AIDCF-001)	Primary 15-domain family	Secondary
D1 Physical Data Centre Controls	Site and perimeter security	PHY-01, PHY-04	—
	Vehicle, loading and delivery	PHY-05, PHY-01	—
	Identity and physical access	PHY-04	SEC-01
	Security zoning and critical areas	PHY-03	—

Legacy domain	Legacy concern (AIDCF-001)	Primary 15-domain family	Secondary
	Surveillance and intrusion detection	PHY-04	—
	Visitor and contractor management	PHY-04	SUP-06
	Asset movement and chain of custody	PHY-05	SUP-03
	Insider-risk safeguards	PHY-04	SEC-01, GOV-02
	Security operations, response, reconstruction	PHY-04	SEC-05, OPS-04
D2 Facilities Engineering	Design basis and engineering authority	GOV-03	CAP-05
	Structural and rack loading	PHY-02	—
	Space planning and maintainability	PHY-02	OPS-03
	Electrical and mechanical architecture	POW-04, THM-01	POW-01, THM-04
	Fire detection and suppression	PHY-06	PHY-03
	Water and leak management	THM-05	PHY-03
	BMS, EPMS and operational technology	SEC-02	POW-06, THM-06
	Commissioning and engineering change	CAP-06, OPS-05	RES-05
D3 Power and Cooling Resilience	Utility capacity and diversity	POW-01	DEP-03
	Distribution, UPS and batteries	POW-03, POW-04	ENV-05
	Standby generation and fuel	POW-02	DEP-03
	Rack power	POW-04	POW-05
	Air and liquid cooling	THM-02, THM-03	—
	CDUs, pumps and heat exchangers	THM-03	THM-04
	Leak and thermal detection	THM-05	THM-02
	Capacity reserve, failover and restoration	THM-01, POW-06	RES-04, CAP-04
	Integrated resilience testing	RES-05	—
D4 Environmental Standards and Resource Performance	Obligations register	GOV-05	ENV-04
	Energy management and PUE	ENV-02	—
	Water use and WUE	ENV-03	THM-04
	Carbon and green-energy claims	ENV-04	—
	Equipment efficiency and compute utilisation	ENV-02	CMP-03
	Higher-temperature operation	ENV-01, ENV-02	THM-01
	Refrigerants and hazardous materials	ENV-05	PHY-06
	Waste, lifecycle and reporting integrity	ENV-05, ENV-04	—
D5 Building Operations	Accountability and procedures	OPS-01, OPS-02	GOV-02
	Shift management and handover	OPS-01	—
	Competency and authorisation	OPS-01	GOV-06

Legacy domain	Legacy concern (AIDCF-001)	Primary 15-domain family	Secondary
	Permit to work	OPS-02	—
	Change and configuration management	OPS-05	—
	Preventive and predictive maintenance	OPS-03	POW-06
	Spares and obsolescence	SUP-05	—
	Vendor governance	SUP-01, SUP-02	—
	Incident, crisis and exercise management	OPS-04, RES-06	RES-05
D6 AI Compute Infrastructure	Compute architecture	CMP-01	—
	GPU and accelerator inventory; ownership and criticality	CMP-01	CAP-01, CAP-03
	Hardware provenance and platform integrity	SUP-03	CMP-04
	Firmware and management controllers	CMP-04	SEC-02, SUP-04
	Cluster configuration and isolation	CMP-02	SEC-03
	Scheduling and fault containment	CMP-03, CMP-02	RES-02
	Capacity and telemetry	CMP-03, CMP-05	—
	Lifecycle and decommissioning	CMP-06	PHY-05
D7 AI Storage and Data Infrastructure	Architecture, inventory and ownership	STO-01	CAP-01
	Capacity, performance and throughput	STO-02	—
	Fault domains and replication	STO-04	RES-02
	Backup and recovery	STO-04, STO-05	RES-04
	Checkpoint and model-artifact storage	STO-05, STO-01	—
	Corruption detection and integrity validation	STO-03	—
	Media handling and sanitisation	STO-06	PHY-05
	Storage telemetry	STO-06	—
D8 AI Network and Interconnect Architecture	Reference topology; compute and storage fabrics; InfiniBand and RoCE	NET-01, NET-03	—
	Management and out-of-band networks	NET-04	SEC-03
	Segmentation and cluster isolation	NET-03	SEC-03
	East-west visibility	NET-06	SEC-05
	Redundancy and congestion management	NET-01, NET-02	—
	Time synchronisation	NET-06	—
	Inter-site and external connectivity	NET-03	DEP-03
	Failure and recovery testing	RES-05	NET-04

45. Domains with no legacy equivalent

Seven of the fifteen domains have no single legacy parent. GOV draws on AIDCF-001 Part X and the D2 engineering-authority and D4 obligations concerns; CAP on AIDCF-001 §15; DEP on AIDCF-001 §17 and §47; RES on AIDCF-001 §19 and §54 and the D3 restoration and testing concerns; SUP on the D5 vendor and spares

concerns and D6 provenance; SEC on D2 OT, D6 firmware and management controllers, D8 segmentation and the AIDCF-001 §57–§59 boundary with AI Trust Graph; EVA on AIDCF-001 Part VII.

> **CCM-REQ-033 Domain model.** New and revised artifacts shall organise content by the fifteen domain codes. D1–D8 shall appear only in legacy-traceability tables.

Part XI – Interpretation rules

46. Rules that bound every AIDCF conclusion

ID	Rule
IR-01 Scope	A conclusion applies only to the declared Capability Instance and assessment boundary. Nothing outside the boundary is assured, and dependencies crossing it are reported with their evidence status.
IR-02 States	A conclusion applies only to the declared and evidenced operating states. Silence about a state is not a claim about it.
IR-03 Time	A conclusion is as of its date and covers the evidence period. It is presumed current only within its validity period and while no Reassessment Trigger has fired.
IR-04 Evidence	A conclusion cannot assert a claim type, scope, population or state wider than its evidence. Sampled results do not extrapolate beyond the sampled population and conditions.
IR-05 Claim type	A conclusion about design adequacy is not a conclusion about implementation, operation, effectiveness or resilience.
IR-06 Non-inheritance	Certificates, third-party reports, vendor attestations and prior assessments are evidence (usually E5 or E1) to be evaluated against the assertion; they do not transfer assurance.
IR-07 UNKNOWN	UNKNOWN is non-favourable and never read as "probably fine" or "probably failed"; its effect is the severity effect, no more and no less.
IR-08 Absence	A claim that something is absent ("no single point of failure", "all dependencies identified") is supported only by documented discovery coverage, method sensitivity and residual UNKNOWN, and is worded as "none identified within documented coverage".
IR-09 Scores	A score is a summary of control results for a stated scope and coverage. It is not a probability, a rating of the facility, or a decision.
IR-10 Maturity	A maturity level describes institutionalisation of practice. It does not indicate that any control is CONFORMING or that any gate PASSes.
IR-11 Conditions	A condition attached to a decision is part of the decision; reliance on the decision without the condition being met is reliance outside its terms.
IR-12 Engineering authority	An AIDCF conclusion does not certify engineering compliance, life safety or legal compliance; those remain with the competent authority.

> **CCM-REQ-034 Interpretation rules.** Reports, statements and tools shall apply interpretation rules IR-01 to IR-12 and shall not word, display or export a conclusion in a way that implies a wider scope, more states, a longer period, a higher claim type or a different authority than the decision supports.

Part XII – Conformance requirements index

47. Index

Requirements are listed in order of appearance. Each is normative where it appears; this index is informative.

ID	Title	Section
CCM-REQ-001	Precedence	3
CCM-REQ-002	Subject ownership	3
CCM-REQ-003	Canonical codes	5
CCM-REQ-004	Public identifiers	5
CCM-REQ-005	Top claim	8
CCM-REQ-006	Single chain	9
CCM-REQ-007	Forward completeness	9
CCM-REQ-008	Reverse traceability	9
CCM-REQ-009	Assurance-case form	10
CCM-REQ-010	Defeaters are retained	10
CCM-REQ-011	Decomposition order	11
CCM-REQ-012	Completeness sub-claims	11
CCM-REQ-013	Undeclared states	11
CCM-REQ-014	Claim bounds	12
CCM-REQ-015	Concept fidelity	13
CCM-REQ-016	Safeguard–control separation	18.2
CCM-REQ-017	Decision authority	21.7
CCM-REQ-018	Materiality test	22.2
CCM-REQ-019	No size proxy	22.2
CCM-REQ-020	Control identity	23.3
CCM-REQ-021	Per-claim-type assessment	23.3
CCM-REQ-022	Claim-type presupposition	23.3
CCM-REQ-023	Vocabulary registry parity	Part VI
CCM-REQ-024	Evidence floor governs result	26
CCM-REQ-025	UNKNOWN classification	31.3
CCM-REQ-026	Gate non-compensation	35.3
CCM-REQ-027	Gate roll-up	35.3
CCM-REQ-028	Ceiling discipline	36
CCM-REQ-029	Conditions	36
CCM-REQ-030	Score presentation	37
CCM-REQ-031	Ladder qualification	39
CCM-REQ-032	No ladder substitution	39

ID	Title	Section
CCM-REQ-033	Domain model	45
CCM-REQ-034	Interpretation rules	46

The series runs consecutively from CCM-REQ-001 to CCM-REQ-034.

Appendix A – Glossary

Terms are listed alphabetically. Where a term is defined in the body, the section is given.

Term	Definition
Absence claim	A claim that something does not exist (for example no single point of failure); supported only by documented discovery coverage and residual UNKNOWN (IR-08).
Accepted risk	A decision-owner acceptance of residual exposure; it never converts NONCONFORMING into CONFORMING.
Action	A recorded remediation or follow-up task with owner, due date and closure evidence (AIDCF-014 `Action`).
AL (Assurance Level)	Programme depth AL0–AL6 (Section 25).
APPROVED_NA	Gate result for a gate not applicable to the declared scope, with approved boundary rationale.
Architecture	Versioned representation of systems, assets, interfaces and resources implementing the Required Functions (Section 15.2).
Assertion	A precise testable proposition to which evidence is attached (Section 19.4).
Assessment	A bounded engagement that evaluates controls and gates for a Decision Question (AIDCF-014 `Assessment`).
Assessment boundary	The declared inclusion perimeter for direct assessment, with explicit exclusions.
Assessment method	One of the ten canonical methods (Section 33.1).
Asset	An identifiable item, service, person, procedure or resource relevant to a Required Function (Section 15.4).
Assumption	A proposition accepted without evidence for the purpose of the argument; material unsupported assumptions become UNKNOWN Records.
Assurance	A reasoned and reviewable level of confidence, for a defined decision, that specified claims are supported by sufficient appropriate evidence within declared scope, states, period and limitations (AIDCF-006).
Assurance case	A structured argument from evidence to a bounded top claim (Section 10).
Assurance Decision	Bounded conclusion issued by accountable authority (Section 21.5).
Assurance Report	Full record of the case and its outcome (Section 21.6).
Assurance Statement	Controlled-wording communication of an Assurance Decision (Section 21.6).
Band withheld	Score presentation when coverage is below 80% or a US4/US5 UNKNOWN exists in scope.
Barrier	A safeguard positioned on a defined failure path (AIDCF-004).
Capability	An intended operational ability supported by infrastructure, as distinct from its instance.
Capability Instance	A specific bounded deployment of an AI capability (Section 14.2).
Capacity Assertion	A qualified statement of a capacity value (Section 15.7).

Term	Definition
Claim type	What a Control Assessment asserts: DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS or RESILIENCE (Section 24).
Common-cause failure	Multiple failures arising from one shared cause.
Common-mode failure	Nominal alternatives failing similarly or through a shared dependency.
Completeness sub-claim	A sub-claim that the set of functions or dependencies identified is sufficient (Section 11).
CONDITIONALLY_ASSURED	Decision state: supported subject to named, owned, time-bound conditions.
Confidence	Reviewer confidence in a claim, CONF-0 to CONF-4 with flags (Section 29).
CONFORMING	Control result: all criteria met for the claim type with evidence at or above floor.
Context	Information within which the argument holds: boundary, baseline, states, exclusions.
Control	A distinct topic-specific normative requirement in AIDCF-007A, assessed per claim type and scope (Sections 18.2, 23).
Control Assessment	Evaluation of one control at one claim type for one scope (Section 20.1).
Control profile tier	Evidence pattern CP-1 to CP-5 (Section 30.3).
Control type	DIRECTIVE, PREVENTIVE, DETECTIVE, CORRECTIVE or RECOVERY (Section 33.2).
Coverage (score)	Scored controls ÷ (in-scope − NOT_APPLICABLE).
Coverage (evidence)	Extent to which evidence represents the required population, period, states and scenarios (dimension C).
CRITICAL (gate role)	A control without whose CONFORMING result at required claim type a gate cannot PASS.
Critical Gate	One of fifteen non-compensating prerequisites CG-01 to CG-15 (Section 35).
Decision lifecycle	DRAFT, RECOMMENDED, APPROVED, ACTIVE, SUSPENDED, WITHDRAWN, SUPERSEDED, EXPIRED (Section 28.2).
Decision Question	The decision an assessment informs (Section 14.1).
Decision Recommendation	Proposed decision state with ceilings, before review (Section 21.4).
Decision state	REASONABLY_ASSURED, CONDITIONALLY_ASSURED, LIMITED, NOT_ASSURED, INCONCLUSIVE (Section 28.1).
Defeater	Evidence, Finding or UNKNOWN that challenges or undermines a claim.
Dependency	Directed reliance of a consumer on a provider (Section 16.1).
Domain	One of fifteen organising areas of the control universe (Section 23.4).
EC level	Evidence-system conformance EC0–EC5 (Section 30.3).
Event	A recorded occurrence (change, incident, test, expiry) evaluated against Reassessment Triggers.
Evidence class	E1–E7 (Section 30.1).
Evidence floor	The minimum evidence (classes, quality, sufficiency) for a result or gate.
Evidence Object	Smallest identified evidence item (Section 19.2).
Evidence Package	Governed collection of Evidence Objects (Section 19.3).
Evidence Requirement	Claim-specific statement of required evidence (Section 19.1).
FAIL	Gate result where a CRITICAL control is NONCONFORMING/PARTIAL or a criterion is demonstrated not met.
Failure Mode	Manner in which a function or element fails (Section 17.3).
Failure Scenario	Ordered, state-qualified path from initiator to consequence (Section 17.4).
Fault Domain	Set of entities affected together by a common cause (Section 16.2).

Term	Definition
Finding	Evidence-supported condition in which a criterion or tolerance is not met or at risk (Section 20.2).
Gate Result	Evaluated status of a Critical Gate (Section 21.2).
GSN	Goal Structuring Notation, a notation for assurance cases.
Hazard	Non-deliberate source of harm or disruption (Section 17.2).
IM level	Implementation Maturity IM0–IM5, diagnostic only (AIDCF-017).
INCONCLUSIVE	Decision state where no determination can be supported.
InfrastructureSystem	Coordinated assets and interfaces providing Required Functions (Section 15.3).
Interface	Defined point of exchange between entities (Section 15.5).
LIMITED	Decision state: some claims supported with material limitations.
MATERIAL	Defined in Section 22.
NONCONFORMING	Control result: criteria materially not met or safeguard demonstrated ineffective.
NOT_APPLICABLE	Control result with approved non-applicability rationale and approver.
NOT_ASSESSED	Control or gate in scope but not yet assessed; not a decision state.
NOT_ASSURED	Decision state where evidence demonstrates the claim or a controlling gate is not met.
Operating envelope	Bounded load, environmental, capacity, duration and state conditions of intended operation.
Operating State	Defined condition of operation (Sections 16.3, 32).
PARTIAL	Control result: some criteria met with deficiencies recorded as Findings.
PASS	Gate result where all CRITICAL controls conform and criteria are met with evidence at floor.
Propagation path	Sequence through which effects move across entities and dependencies.
Quality band	Q0–Q5 overall usability of an Evidence Object for an assertion (Section 30.2).
REASONABLY_ASSURED	Decision state: bounded top claim supported, gates pass, no undisposed US3+ UNKNOWN.
Reassessment Trigger	Declared condition that invalidates reliance and requires reassessment (Section 21.7).
Redundancy	Additional capacity or paths; not by itself evidence of resilience.
Required Function	Function whose delivery is necessary for a Service Outcome in a state (Section 15.1).
Residual state	The state remaining after the safeguards on a path have acted.
Resilience	Demonstrated ability to continue, contain, adapt and recover within tolerance.
Resource	Consumable or constrained input, including time (Section 15.6).
Safeguard	Real-world measure acting on a failure path (Section 18.1).
Score Snapshot	Immutable record of score inputs, outputs and lineage (Section 21.3).
Service Outcome	Required measurable result of a Capability Instance (Section 14.3).
Sufficiency	S0–S5 judgement of whether a package supports a claim (Section 30.2).
SUPPORTING (gate role)	A control that informs but does not automatically decide a gate.
Threat	Deliberate source of adverse consequence (Section 17.1).
Tolerance	Accepted bounded deviation from an outcome target (Section 14.4).
Top claim	The single root claim of an assurance case (Section 8).
UNKNOWN (control/gate result)	Result that cannot be determined from available evidence.
UNKNOWN Record	First-class record of an unresolved material uncertainty (Section 20.3).

Term	Definition
UNKNOWN severity	US1–US5 decision effect of an UNKNOWN (Section 31.2).
UNKNOWN type	U1–U10 reason for an UNKNOWN (Section 31.1).

Appendix B – AIDCF-014 class coverage

Every one of the 48 canonical AIDCF-014 classes is anchored to a concept in this document.

AIDCF-014 class	Concept / section
ReleaseManifest	Release control of registries and artifacts (Part IX)
CapabilityInstance	Section 14.2
ServiceOutcome	Sections 14.3–14.4
RequiredFunction	Section 15.1
ArchitectureSnapshot	Section 15.2
InfrastructureSystem	Section 15.3
AssetInstance	Section 15.4
Interface	Section 15.5
Resource	Section 15.6
Dependency	Section 16.1
FaultDomain	Section 16.2
OperatingState	Sections 16.3, 32
CapacityAssertion	Section 15.7
Threat	Section 17.1
Hazard	Section 17.2
FailureMode	Section 17.3
FailureScenario	Section 17.4
Safeguard	Section 18.1
ControlDomain	Section 23.4
ControlFamily	Section 23.4
Control	Sections 18.2, 23
EvidenceRequirement	Section 19.1
EvidenceObject	Section 19.2
EvidencePackage	Section 19.3
AssuranceProgram	Section 14.1 (Decision Question); AIDCF-011
Assessment	Sections 14.1, 20.1
AssessmentProcedure	Section 20.1
Sample	Section 20.1
TestRecord	Section 20.1
ControlAssessment	Section 20.1

AIDCF-014 class	Concept / section
Finding	Section 20.2
UnknownRecord	Sections 20.3, 31
CriticalGate	Sections 21.1, 35
GateResult	Sections 21.2, 35.3
ScoreSnapshot	Sections 21.3, 37
DecisionRecommendation	Sections 21.4, 36
ReviewRecord	Section 21.5
AssuranceDecision	Sections 21.5, 28
AssuranceStatement	Section 21.6
AssuranceReport	Section 21.6
Action	Section 20.2
ReassessmentTrigger	Section 21.7
Event	Sections 9.2, 21.7
IntegrationHandoff	Section 10 (away goals); AIDCF-005
ValidationResult	Conformance of datasets (Section 5); AIDCF-014
MigrationRecord	Legacy alias migration (Part VI, Section 41)
DistributionControl	Disclosure of reports and sensitive paths; AIDCF-012, AIDCF-014

Final doctrine. One chain, one vocabulary, one meaning per word. Claims are bounded; evidence is specific; gates do not average; UNKNOWN does not disappear; people decide.